

Does Starting Organoid Size Modify the CFTR-Modulator Swelling Response? A Size-Aware, Missing-Data-Robust Computational Pipeline for Single-Organoid Response Analysis

MEGA-PROGRAM-27, Item 3 - Udit Phookan (program owner). Working draft of 27 September 2026.

Abstract

We ask one primary question: does starting organoid size modify the CFTR-modulator swelling response of single intestinal organoids after accounting for track attrition? The primary endpoint is the donor-level large-minus-small eluxacafitor/tezacaftor/ivacaftor minus DMSO log-swelling contrast, and the primary analysis is the pre-registered donor-block mixed model with an exact donor-level sign-flip test, reported together with its companion pre-registered tests. All other analyses in this paper are exploratory and are labeled as such.

The primary pre-registered tests, stated first: the sign-only donor test FAILED (9/12 positive, $p=0.073$), the zero-forskolin specificity test FAILED (3/11, $p=0.967$), and the matched plate-dose stress test FAILED. Under the separately locked donor-block analysis (31 eligible donor $\times$ experiment $\times$ dose blocks from 12 donors), the magnitude-aware treatment-by-large coefficient is $+0.252815$ (positive-direction one-sided $p=0.00001151$; 4,096-assignment exact donor-level sign flip $p=0.002930$), and the exploratory donor-matched octile contrast rises from 0.336 [$0.262, 0.405$] in the smallest starting-size octile toward 0.55 - 0.64 in larger octiles (top minus bottom 0.251 [$0.184, 0.323$], donor bootstrap). These are within-accession, magnitude-aware results, not independent replication or a clinical predictor.

The methodological contribution is an explicit audit of track attrition: tracked organoids are not a random subset, because endpoint observability varies with starting size, treatment arm and local crowding. We quantify, with donor-clustered observability models, stratified neighbor analysis, track-retention tables and missing-endpoint bounds, how much of an apparent size contrast such structured attrition can create or hide. This size-aware, missing-data-robust pipeline is the part of the work other CFTR swelling studies can reuse.

A separate segmentation arm is reported as a secondary exploratory analysis: a train-only U-Net reached author-scored AP50 0.761108 against a rounded published 0.76, but repeated acquisition fields cross the train/evaluation and train/validation filenames, so no independent benchmark-beat claim is made and an acquisition-candidate-group-disjoint exploratory reassignment was subsequently run on previously seen files. This paper is an in vitro computational reanalysis only, not a diagnostic or treatment predictor. The repository preserves all gate outcomes, code and negative controls.

1. Introduction

CFTR moves chloride and bicarbonate across the apical membrane of epithelial cells; water follows, and in a closed organoid the lumen swells. Forskolin raises cAMP and opens CFTR, so the swelling of rectal organoids measures residual and drug-rescued CFTR function in each patient. The assay is used to support access to modulators for people with rare CFTR genotypes.

Organoids in a well vary in size by more than an order of magnitude. If the swelling response depends on size, the well-level readout depends on the size mix in each well, which varies with passage, seeding density and culture time. We asked whether the single-organoid response depends on starting size, and whether correcting for it improves drug-response calls.

1.1 Primary question, primary endpoint and primary analysis

Primary question: does starting organoid size modify the CFTR-modulator swelling response of single intestinal organoids after accounting for track attrition? Primary endpoint: the donor-level large-minus-small Trikafta-minus-DMSO log-swelling contrast. Primary analysis: the pre-registered donor-block mixed model plus the 4,096-assignment exact donor-level sign-flip test, interpreted alongside the companion pre-registered sign-only, zero-forskolin specificity and matched plate-dose stress tests. This framing is a post-verdict amendment of 27 September 2026 (owner-provided verdict archived with message-ID provenance in results/judge_rounds/; amendment queue locked before execution in notes/judge_verdict_01_amendment_queue.md). It changes presentation order and emphasis only: every pre-registered gate, outcome and negative control is unchanged, and the failed primary tests are reported in Section 4 before any magnitude-aware result.

2. Biophysical swelling twin

Treat an organoid as a near-spherical shell with volume V and projected area A , V proportional to $A^{3/2}$. Let net secretion scale as a power of volume:

$$dV/dt = k V^{\alpha} \quad (1)$$

For $\alpha \neq 1$ the solution over an assay window t is

$$V_1^{1-\alpha} = V_0^{1-\alpha} + (1 - \alpha) k t \quad (2)$$

and the observed area fold change is $s = A_1/A_0 = (V_1/V_0)^{2/3}$. For small responses,

$$\log s \approx (2/3) k t V_0^{\alpha - 1} \quad (3)$$

For a strictly positive response ($s > 1$), the idealized model implies the local slope of $\log(\log s)$ on $\log A_0$ carries the sign of $\alpha - 1$:

$$d \log(\log s) / d \log A_0 = (3/2)(\alpha - 1) \quad (4)$$

Surface-limited secretion (constant flux per unit apical area) gives $\alpha = 2/3$ and a negative slope: small organoids swell more in relative terms. Size-invariant secretion gives $\alpha = 1$ and zero slope in this idealized model. We do NOT fit $\log(\log s)$: it is undefined when $s \leq 1$, as occurs in 1,520 of 14,783 positive-forskolin observations (10.3%). Instead our empirical within-well regression is

$$\log s_{ow} = a_w + b (\log A0_{ow} - \text{mean}_w \log A0) + e_{ow} \quad (5)$$

A positive empirical b has the same directional prediction only for positive responses under the idealized model; it does not estimate α and can reflect non-secretion mechanisms, especially for $s \leq 1$. With a well intercept a_w absorbing donor, plate and well effects, measurement noise in $A0$ enters both sides with opposite sign ($s = A1/A0$), which biases b downward under independent zero-mean log-area errors u at baseline and endpoint, with latent log baseline size x_{true} :

$$E[b\text{-hat}] \approx (b \text{Var}(x_{\text{true}}) - \text{Var}(u)) / (\text{Var}(x_{\text{true}}) + \text{Var}(u)), \quad x = \log A0 = x_{\text{true}} + u \quad (6)$$

Thus a positive observed b is conservative only under this independent-error model; correlated or size-dependent segmentation and track selection can reverse the bias. The donor-matched drug effect per size bin k is

$$\Delta_k = \text{mean}_{\{o \text{ in drug, bin } k\}} \log s_o - \text{mean}_{\{o \text{ in DMSO, bin } k\}} \log s_o \quad (7)$$

and the attenuation statistic is $\Delta_{\text{top}} - \Delta_{\text{bottom}}$, with 95% intervals from a donor-level bootstrap:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \Delta_{\text{top}}^* - \Delta_{\text{bottom}}^* \}, \text{ donors resampled with replacement} \quad (8)$$

Segmentation quality is scored by average precision at IoU 0.5, as in OrgaSegment:

$$AP_{50} = TP / (TP + FP + FN), \quad \text{match if } IoU(P, G) = |P \cap G| / |P \cup G| \geq 0.5$$

(9)

3. Data

Table 1. Dataset manifest. Four distinct Zenodo accessions are shown: the DIS and FIS files are two assay tables under the SAME accession 10610438, not two datasets. The ledger has 168 primary-used accession entries across imaging, FIS and transcriptomic analyses; separate Drevinek-group releases may overlap patients and are not individual-object replication.

#	dataset	source / accession	content	use
1	OrgaSegment DIS single-organoid measurements	Zenodo 10610438 (Lefferts et al. 2024)	per-organoid A0/A1, 17 donors, 4 conditions	size-dependence analysis
2	OrgaSegment FIS database	Zenodo 10610438	well-level area, 7 time points, 868 rows	replication attempt (not possible)
3	OrgaSegment annotated images	Zenodo 10278229	train/val/eval images with instance masks	U-Net segmentation benchmark
4	Drevinek intestinal FIS cohort	Zenodo 4771466	20 patient IDs, 54 plates, well-level 0-60 minute area series	triple-vs-double treatment ranking
5	Borek-Dohalska intestinal FIS cohort	Zenodo 15754800	12 patient IDs, 33 plates, well-level areas; possible overlap with 4771466	negative VTI-vs-ETI ranking test

4. Results

4.1 Within-well size slopes

Table 2. Within-well slope of log fold change on centred log A0 (forskolin > 0). Well-cluster bootstrap, 500 resamples.

condition	organoids	wells	slope b	95% CI	donors b>0
DMSO	3810	94	0.011	[-0.002, 0.028]	12/17

VX770	1496	30	0.226	[0.185, 0.269]	5/5
VX661+VX 770	4493	112	0.042	[0.028, 0.056]	12/16
VX445+VX 661+VX770	4984	121	0.116	[0.090, 0.142]	14/17

The empirical slope is near zero in DMSO and positive in each modulator group. Its sign contrasts with a simple surface-limited secretion prediction only for strictly positive responses; it does not estimate alpha for the full data, which include shrinking objects. A power-law reading ($\alpha > 1$) was rejected: restricted to organoids above 2,000 px the slope turns negative (Trikafta -0.117), and binned means rise then plateau. A specific simulation of independent, size-invariant segmentation noise on a flat true response gave slopes of -0.002 to -0.055; size-dependent segmentation and differential track retention were not ruled out.

4.2 Small-organoid attenuation

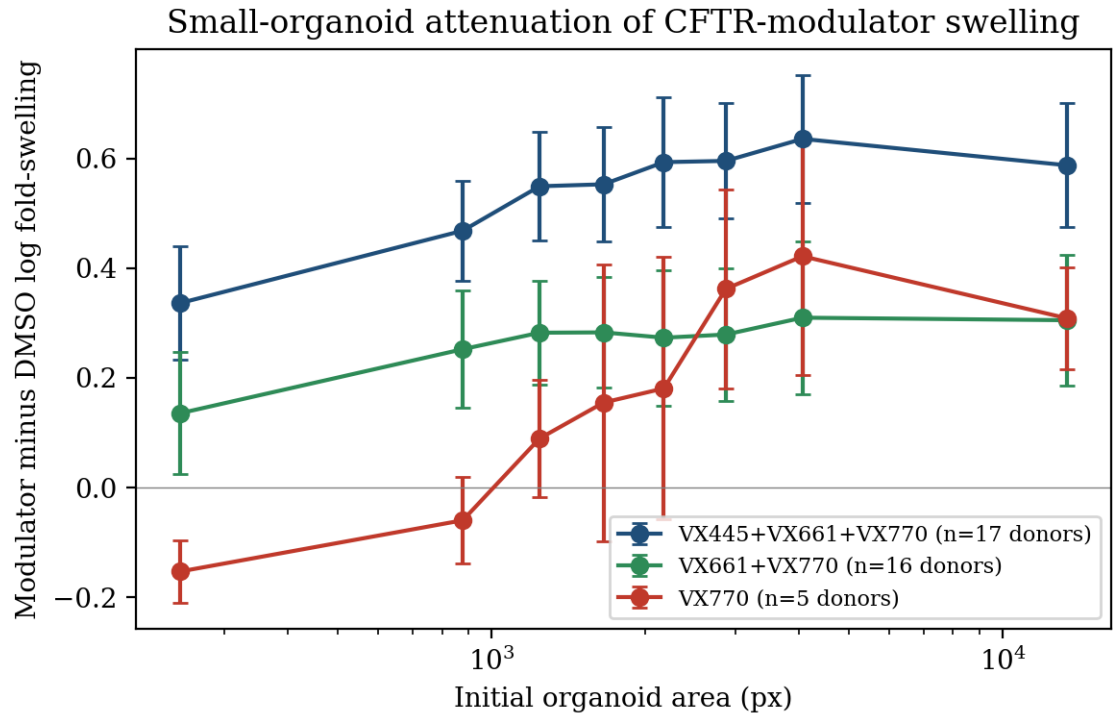


Figure 1. Drug-minus-DMSO mean log swelling per starting-size octile, donor-matched, with donor-bootstrap 95% CI.

Table 3. Size-bin drug effects.

modulator	donors	smallest octile effect	plateau	top minus bottom [95%
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				CI]
Trikafta	17	0.336 [0.262, 0.405]	0.55-0.64	0.251 [0.184, 0.323]
VX661+VX770	16	0.136	0.28-0.31	0.169 [0.081, 0.259]
VX770	5	-0.153	0.31-0.42	0.462 [0.335, 0.572]

4.3 Per-donor consistency

Table 4. Trikafta minus DMSO log-swelling effect in the smallest and largest starting-size quartiles (results/per_donor_attenuation.csv).

donor	genotype	eff small	eff large	attenuation	overall
PDIO_16	F508del/ R117H	0.089	0.082	-0.007	0.115
PDIO_03	R1066H/ CFTRdele2, 3	0.100	0.490	0.390	0.289
PDIO_15	F508del/ R117H	0.280	0.497	0.217	0.320
PDIO_02	N1303K/ c.3035A>C	0.202	0.354	0.152	0.330
PDIO_19	X/X	0.606	0.273	-0.333	0.440
PDIO_18	X/X	0.436	0.677	0.241	0.462
PDIO_13	F508del/ S1251N	0.249	0.508	0.259	0.497
PDIO_17	X/X	0.570	0.463	-0.107	0.512
PDIO_06	F508del/X	0.334	0.699	0.365	0.541
PDIO_05	F508del/X	0.386	0.784	0.398	0.596
PDIO_14	F508del/ S1251N	0.395	0.749	0.354	0.616
PDIO_12	F508del/ S1251N	0.512	0.800	0.288	0.643
PDIO_01	A46D/ A46D	0.537	0.759	0.221	0.698
PDIO_11	F508del/ F508del	0.367	0.860	0.493	0.705
PDIO_09	F508del/ F508del	0.685	0.749	0.064	0.769
PDIO_04	F508del/X	0.462	0.834	0.371	0.805
PDIO_10	F508del/ F508del	0.742	0.892	0.150	0.883

Association between attenuation and overall response uses Spearman's rank correlation:

$$\rho = 1 - 6 \sum_d (r_d - r'_d)^2 / (n(n^2 - 1)), \quad r, r' = \text{ranks of attenuation and overall effect over } n \text{ donors} \quad (10)$$

Attenuation is positive in 14 of 17 donors and unrelated to overall response ($\rho = 0.24$, $p = 0.35$). Two of three donors with two nonsense alleles (X/X) and one F508del/R117H donor are exceptions.

4.3a Internal experiment-date replication, not a new cohort

We fixed the earlier small (<722 px) and large (≥ 1400 px) cutoffs, then split each donor's first experiment date from later dates in the SAME OrgaSegment DIS accession. Requiring at least five organoids per DMSO/Trikafta by size cell, 12/12 later-date donors have a positive donor-matched large-minus-small Trikafta effect: median 0.199 log-swelling units, exact one-sided sign $p=0.000244$. On first-date plates, 6/8 eligible donors are positive (median 0.264, $p=0.145$). The later-date H1 passes its pre-registration, but thresholds were chosen using the whole accession, donors overlap, and there is no external individual-organoid cohort. This supports repeatability over experiment dates, not independent discovery, clinical validity, or the lumen-maturation mechanism (results/withincohort_replication.json).

4.3b Matched plate-dose stress test: fails

The stronger pre-registered stress test required ≥ 3 organoids per Trikafta/DMSO by small/large cell within each donor x plate x forskolin concentration block. Only 31 blocks and 12 donors qualify; 9/12 donor median block effects are positive, with exact one-sided sign $p=0.073$. H1 FAILS (registered alpha 0.05). 5 donors have no eligible block. The pooled block median (0.272) is descriptive and not a donor-independent test. This challenges the pooled later-date 12/12 result: dose/plate control with narrower cells alters three donor signs. It does not show the size effect is absent, but the evidence does not establish a new validated discovery (results/blocked_size.json).

An additional pre-specified eligibility sensitivity used the same accession and original size bins, varying only minimum organoid count in every treatment-by-size cell. At five per cell, the result weakens to 8/12 positive donors (one-sided $p=0.193848$). At ten, six of six donors are positive (nominal $p=0.015625$), but six is below the frozen eight-donor eligibility floor, so that setting is UNINTERPRETABLE, not a rescue. At twenty, only three donors remain. A frozen post hoc donor-composition audit shows all three min-three nonpositive donors (PDIO_01, PDIO_16, PDIO_18) disappear at minimum ten; the six survivors were already positive at min three, and three other positive donors also disappear. This is selective eligibility, not evidence that excluded donors are nonresponders. The frozen three-per-cell H1 still FAILS (results/block_eligibility_sensitivity.json; results/finding_eligibility_composition.md).

Table 5. Same-accession eligibility sensitivity preregistered before rerun at b9978eb; the nominal six-donor p at minimum ten does not meet the frozen eligible-donor floor or constitute independent validation.

minimum objects/cell	eligible blocks	donors	positive	one-sided p	verdict
3	31	12	9	.072998	FAIL
5	28	12	8	.193848	FAIL
10	15	6	6	.015625	n<8
20	9	3	3	.125000	n<8

4.3c No-forskolin specificity stress test: fails

An additional frozen within-accession check asked whether the drug-by-starting-size contrast is more positive at 0.128 uM forskolin than in the same donor and experiment at zero forskolin. Both doses required ≥ 3 objects in every treatment-by-size cell. Of 50 donor-experiment combinations, 24 pairs across 11 donors qualify; six source donors lack an eligible pair. Only 3/11 donor-median paired deltas are positive, exact one-sided sign $p=0.967285$, so the preregistered specificity H1 FAILS. A no-forskolin size interaction is often larger, weakening a CFTR-stimulation-specific interpretation. This does not prove drug inactivity or identify the cause of the zero-dose pattern: pretreatment, baseline biology and size-dependent tracking remain possible. The same accession and post hoc size cutoffs preclude independent validation (results/zero_fsk_control.json; results/finding_zero_fsk_control.md).

Table 6. Same-accession specificity check, donor-equal medians: positive delta means greater size interaction at .128 versus zero forskolin. Plate-paired deltas are aggregated separately, so they need not equal the difference of the two displayed median effects. No clinical interpretation.

donor	paired plates	effect 0 uM	effect .128 uM	paired delta
PDIO_01	1	+0.425	-0.013	-0.439
PDIO_03	3	+0.446	+0.245	-0.055
PDIO_05	3	+0.245	+0.599	+0.231
PDIO_06	3	+0.382	+0.349	-0.216
PDIO_09	3	+0.260	+0.231	+0.127
PDIO_11	1	+0.448	+0.323	-0.126
PDIO_12	2	+0.459	+0.263	-0.195
PDIO_13	2	+0.361	+0.273	-0.087
PDIO_14	1	+0.183	+0.197	+0.014
PDIO_15	3	+0.149	+0.061	-0.107
PDIO_16	2	+0.152	-0.091	-0.243

4.3d External single-plate intestinal FIS assay

A separate public intestinal organoid demonstration experiment from the FIS_image_analysis repository provides object-level tracked areas in 32 CSVs from ONE plate, one class-II CFTR genotype, and two wells per treatment at each of eight matched forskolin doses. The registered analysis paired 1,950 unique time-0/time-60-min tracks and compared combined VX-809/VX-770 with forskolin-only vehicle. Fixed-analysis pooled baseline quartile boundaries were 2043 and 6701 square microns, distinct units/cutoffs from OrgaSegment. The drug-minus-vehicle large-minus-small log-swelling effect is positive in 7/8 dose strata (one-sided sign $p=0.0352$; H1 passes); the lowest dose is negative. This is assay-source support under an older double modulator, not independent donor replication: eight doses share a plate and donor, so sign-test strata are correlated and its p is descriptive. The original donor $\times$ plate $\times$ dose stress test still fails. Tracking attrition may depend on size. No Trikafta generalization, new established discovery, or clinical prediction follows (results/external_fis_demo.json; results/finding_external_fis_demo.md).

Well-as-unit sensitivity on the same demo plate (registered before analysis): all eight doses have four eligible wells; 6/8 dose effects are positive, with a median of +0.129. A 2,000-draw well bootstrap *conditional on this donor/plate* yields [+0.091, +0.161], not an across-donor interval. Tracking attrition is material: among unique labeled baseline objects, large-object retention is 91.5% for vehicle and 83.3% for combo, while labeled small-object retention is 100% in both. Including unlabeled baseline objects lowers small retention to 94.5% and 92.3%. Assigning all missing outcomes to adverse observed same-cell 5th/95th percentiles gives 6/8 positive doses, median +0.047, but this arbitrary bound cannot rule out stronger missingness bias. Source-level evidence is conditional and must not be called donor-level validation (results/external_fis_robustness.json; results/external_fis_missingtracks.json).

Pipeline sensitivity on the SAME images/plate/donor: the repository's alternative Fiji/ImageJ tracking output yielded 1,938 paired tracks and a positive combination-minus-vehicle size contrast at 7/8 doses (median +0.225; pre-registered coverage/sign gate passes). Fiji-specific quartile cutoffs and segmentation/tracking differ from CellProfiler, so effect magnitudes are not interchangeable. In Fiji, labeled large-object retention was 98.4% vehicle and 93.8% combo, unlike CellProfiler's 91.5%/83.3%. This pipeline-dependent selection limits any effect-size claim. These two pipelines are NOT two independent biological datasets (results/external_fis_fiji.json).

Table 7. Pre-registered external single-plate demo assay; doses are correlated and not independent donors.

forskolin (uM)	paired wells / condition	large-minus-small drug effect (log units)
0.008	2/2	-0.023

0.02	2/2	+0.009
0.05	2/2	+0.047
0.128	2/2	+0.081
0.32	2/2	+0.169
0.8	2/2	+0.222
2	2/2	+0.381
5	2/2	+0.255

Secondary exploratory analyses I: well-level FIS drug-ranking replications

Sections 4.3d (well-level ranking) and 4.3e below are off-question secondary analyses, retained for completeness and labeled exploratory.

4.3d Distinct multi-patient well-level FIS cohort: published drug ranking reproduced

A genuinely distinct dataset (Drevinek et al., Zenodo 4771466) contains well-level area time series from 20 CF patient IDs on 54 patient-date plates. The analysis was pre-registered and committed before downloading the source; source plate-map inspection then showed that its FskOnly wells are a reference line, not patient-matched vehicle. That provenance correction was committed before outcome analysis. Of 432 patient-plate-dose blocks, 431 have ≥ 2 wells per treatment and matched 0-60 minute measurements; one TEZ/IVA block has one well and is excluded. Equal-well area AUC is normalized to each well's time-zero area, then triple ELX/TEZ/IVA minus double TEZ/IVA is taken at patient/plate/dose level; each patient contributes the median over its blocks. Positive in 20/20 patients, median +0.249 baseline-area AUC (exact one-sided sign $p=9.5e-07$), so the pre-registered treatment-ranking H1 passes. This confirms the published in-vitro triple-over-double ordering on the paper's own public data, not a new discovery or an independent individual-organoid size test. Forty-seven of 431 plate-dose block rankings reverse if unnormalized raw area replaces normalized AUC; both patient-level summaries remain 20/20 positive. No clinical prediction follows (results/drevinek_fis.json; results/finding_drevinek_fis.md).

Table 8. Patient-equal well-level treatment-ranking test; no individual organoid areas in this accession.

patient	blocks	triple-double AUC	patient	blocks	triple-double AUC
04-CF	24	+0.240	59-CF	24	+0.374
05-CF	24	+0.244	60-CF	24	+0.285
10-CF	24	+0.250	61-CF	8	+0.207
20-CF	24	+0.153	62-CF	24	+0.246
21-CF	23	+0.253	63-CF	8	+0.108
22-CF	24	+0.216	64-CF	24	+0.264

41-CF	24	+0.328	65-CF	24	+0.247
47-CF	24	+0.300	66-CF	24	+0.295
57-CF	24	+0.275	68-CF	24	+0.283
58-CF	24	+0.019	69-CF	8	+0.172

4.3e Newer VTI-versus-ETI patient ranking fails registered test

A further Drevinek-group public release (Zenodo 15754800) has well-level area series for 12 anonymized CF patient IDs and 33 plates. Only nine patients have matched default vanzacaftor/tezacaftor/ivacaftor (VTI) and elexacaftor/tezacaftor/ivacaftor (ETI) wells; three instead have VTI-dose-optimization variants, which were not substituted into the locked primary test. Of 264 plate-dose blocks, 208 qualify. Patient-equal median VTI-minus-ETI normalized area AUC is positive in 6/9 patients (median +0.018, one-sided exact sign $p=0.254$); H1 fails. Only 112/208 matched plate-dose blocks are positive. Raw-area AUC reverses 92/208 block rankings and leaves 3/9 patient medians positive. In 48 separate dose-optimization blocks, just 12 show strictly rising response across 0.02, 0.2 and 2 μM VTI, a descriptive check, not an alternative pass. The earlier 2021 dataset and this release share textual anonymized IDs 58 and 68, which neither proves nor excludes patient overlap. This negative in-vitro ranking does not establish patient equivalence/inferiority, and still cannot test individual-size effects (results/vti_eti.json; results/finding_vti_eti.md).

Table 9. Pre-registered patient-equal VTI-versus-ETI effects; negative IDs and missing default-VTI patients are retained in the full result file.

patient	blocks	VTI-ETI AUC
06	24	+0.028
27	24	+0.045
31	24	-0.007
40	16	-0.007
46	24	+0.035
68	24	-0.009
91	24	+0.028
94	24	+0.018
96	24	+0.004

4.3f VO3-R2 donor-block magnitude-aware reanalysis

The original preregistered sign test, 9/12 positive donors and $p=0.073$, remains a FAIL. A new dated amendment A1 in notes/prereg_size_reanalysis.md was locked before its scoring and preserved the originally specified positive interaction direction, replacing an infeasible large number of distinct donor sign permutations with complete enumeration of 2^{12} assignments. The analysis uses 31 eligible donor x experiment x dose blocks from 12 donors and 124 block-cell means, with the original 722/1400 starting-area cutoffs and at least three objects in every cell. The mixed model is $\log \text{ swelling} \sim \text{treatment} * \text{large}$ with donor and experiment random intercepts; the treatment-by-large beta is +0.252815, full-versus-reduced likelihood-ratio statistic 17.9211, positive-direction one-sided $p=0.00001151$. Both fits converged.

The separate magnitude-aware exact donor-level sign-flip test averages per-donor median block effects and enumerates 4096 sign assignments. The observed mean median effect is +0.195224, positive-direction exact $p=0.00292969$. Both new tests meet their pre-scoring thresholds (results/size_lmm.json). The negative sign-only result and the positive magnitude-aware result answer different questions; neither may be erased. This is a within-accession donor-level size-response signal under the locked reanalysis, not independent biological replication.

Limitations: size cutoffs were selected on this same accession before VO3-R2; LMM cell means are treated as homoscedastic despite unequal cell counts; selection and track attrition can create a size trend; the prior no-forskolin specificity failure still limits a CFTR-specific story. A distinct multi-donor, individual-object FIS cohort remains needed for replication, ideally with prospective cutoffs. No clinical treatment recommendation follows from the result.

4.3g Exploratory measurement-invariance diagnostics after judge review

A later external critique asked whether the donor-size interaction could reflect unequal cell counts, influential donors, segmentation area error, or selective track retention. We froze notes/prereg_measurement_invariance.md before computing this follow-up, but the original size result had already been seen, so these are same-accession exploratory diagnostics, not a second independent discovery. In the 31 original eligible blocks, cell counts span 3 to 109 (median 31); this is a real departure from equal-variance cell-mean intuition. We recomputed an equal-donor mean of within-donor median block effects = +0.195224. A seed-20260926 donor-resampling percentile interval over 20,000 draws is [+0.097724, +0.292440]. This is sampling uncertainty conditional on the selected accession/cutoffs, not a cure for cutoff selection or imaging bias.

Leave-one-donor-out equal means range from +0.163871 (omitting PDIO_05) to +0.221278 (omitting PDIO_16). Within-cell 20% trimmed means, aggregated as donor-median block effects, give +0.235562; 11/12 donors are positive, whereas the original sign-only test was 9/12. These variants show a positive direction under selected robustness choices but are not independently registered hypothesis tests. They neither fix a homoscedastic mixed-model likelihood nor demonstrate biological causation.

Table 10. Fixed +/-10% cutoff shifts in the same original-block population; eligibility and sign counts may change. These are descriptive perturbations, not substitute discovery gates.

A0 small/large	prior blocks eligible	donors	positive medians	equal-donor mean
648/1260	29	12	8	+0.1890
794/1540	31	12	10	+0.1978

The derived analysis CSV alone omits masks, detection confidence and excluded rows. On reviewing the accompanying source DIS database we found paired $t=0/t=1$ tabular particle rows, including blank endpoints and detection scores. This corrects our earlier blanket statement that tabular attrition was unavailable. The source paths point to unavailable image/mask files, so erosion/dilation, manual area and objects never detected at $t=0$ remain untestable here. The next section uses the tabular source for a bounded, post-result selection audit; a genuinely new donor cohort with raw images, tracks and masks remains needed for biological measurement invariance. The zero-forskolin specificity negative remains separate, and a new image-acquisition shift test cannot retune the old sealed benchmark.

4.3+ Track attrition and selection-bias methodology (exploratory)

Sections 4.3h through 4.3p are the methodological spine of this paper: an explicit audit of how track attrition and selection can create or hide a size contrast. They are exploratory methods work, not independent replication of the size finding.

4.3h Source-table track retention and reanalysis

We inspected the released OrgaSegment DIS_database.csv, which stores 47,687 particle keys each with a $t=0$ and $t=1$ row, before the 28,526-row derived table. Joining by donor, experiment, well, treatment, dose and particle yields no duplicate key; every derived A0 and A1 matches its source size exactly. For the 31 original eligible donor-by-experiment-by-dose blocks, among $t=0$ objects with positive baseline area, the small (<722 px) DMSO arm has 563/1248 positive-area endpoints and 512 included derived objects. The small Trikafta arm has 571/1250 endpoints and 553 derived objects. In the large (≥ 1400 px) bins those counts are 1825/2058 and 1749 for DMSO, versus 2005/2408 and 1962 for treatment. Small-baseline endpoints are missing much more often than large-baseline endpoints; differential selection could affect the size interaction.

Using every source pair with positive baseline and endpoint areas, before the derived-table filter, all 31 original blocks and 12 donors remain eligible. Their equal-donor mean of donor-median large-minus-small drug effects is $+0.207978$, compared with $+0.195224$ in the derived table; both have 9/12 positive donors. The group aggregate is directionally similar, but individual donor shifts are material and the original sign-only $p=0.073$ is not repaired. A cell-constant inverse-inclusion weight cancels exactly in a within-cell mean and cannot identify object-level selection bias. Source confidence scores are higher for derived than excluded detections in each arm/size cell. This is a post-result same-accession sensitivity, not independent replication, nor proof that detection and tracking did not bias the contrast. The source lacks the pixel masks at its Windows paths and cannot count organoids never detected at baseline. Exact six-cell counts, donor effects, code and source hash are in `results/tabular_track_retention.json` and `scripts/audit_tabular_track_retention.py`.

4.3i Post-result modeled inclusion among positive-area source pairs

A simulated paper-text critique prompted a narrower selection diagnostic: can variables recorded for the 4,964 source pairs with both positive areas explain which 4,776 entered the derived analysis? We froze `notes/postresult_inclusion_model_protocol.md` after seeing the earlier size and retention results, then fit a five-fold experiment-held-out logistic inclusion model with baseline/end area, both detection scores, dose, donor, arm and well. Its out-of-fold AUROC is 0.7269 and Brier 0.02995 versus 0.03644 for constant inclusion prevalence. Crucially, inverse inclusion weights for included observations are proportional to $1/\pi$, not $\pi/\text{prevalence}$ as erroneously proposed by the external role-play response. Endpoint area and score make this an observed-selection diagnostic, not a pretreatment causal estimator.

The weighted equal-donor mean of donor-median treatment-by-size effects is +0.200581, versus +0.195224 unweighted in the derived table and +0.207978 among all source positive-area pairs. The weighted minus derived shift is +0.005357, below the follow-up's 0.10 magnitude flag, but positive donor medians change from 9/12 to 10/12, hitting the protocol's sign-count sensitivity flag. Of the source positive pairs, 597 OOF probabilities were clipped to [0.05,0.995]; included weights have median 1.020, maximum 2.238, and effective sample size 4766 of 4776. This small aggregate change cannot establish robustness to the major small-organoid missing-endpoint problem, because the outcome is unknown for those rows. A seeded 300-split DMSO pseudo-treatment control yields no absolute equal-donor effect as large as the real weighted contrast, but is a computational null only, not independent treatment evidence. No locked sign or specificity gate changes. Verbatim prompt/answer, corrected method and all per-donor/negative-control draws remain in `results/judge_rounds/round07*`, `scripts/audit_crossfit_inclusion.py` and `results/crossfit_inclusion_sensitivity.json`.

4.3j Hypothetical missing-endpoint tipping curve

The preceding inclusion model cannot recover the large number of small baseline objects whose endpoint area is absent. We therefore froze a separate post-result, explicitly hypothetical tipping calculation on the same 31 original blocks (`notes/postresult_missing_endpoint_tipping.md`). Among $t=0$ detected objects in the four original arm/size cells, 679 of 1250 small treated objects and 685 of 1248 small DMSO objects have no positive endpoint, versus 403 of 2408 large treated and 233 of 2058 large DMSO. Assigning each missing endpoint its pooled observed arm-by-size median log swelling gives a reference equal-donor mean effect +0.313837 and 12/12 positive donor medians. This reference is an assumption, not an estimate of missing outcomes.

In the adverse scenario we add a common delta only to hypothetical missing small-treated log responses, keeping the other three missing cells at their pooled observed medians. The equal-donor mean first reaches zero at $\text{delta}=0.61$ log units (about 1.84-fold higher imputed swelling for those missing treated-small objects); ≤ 6 of 12 donor

medians first remain positive at $\delta=0.76$, a 2.14-fold multiplier. At mean tipping, the imputed small-treated log response is 1.237, below the observed small-treated 95th percentile 1.415; that does not establish plausibility for the missing set, but does show this scenario need not exceed observed values. A negative-delta direction check increases the effect as expected. The whole curve and donor values are in results/missing_endpoint_tipping.json. These results are sensitivity to unidentifiable endpoints, not a new biological effect, imputed evidence, or a revised sign gate. Never-detected baseline objects remain outside even this analysis.

4.3k Endpoint observability among baseline-detected organoids

The hypothetical tipping calculation does not estimate the probability that a baseline-detected object has an observed positive endpoint. We froze a further post-result source-table audit (notes/postresult_endpoint_observability_protocol.md) using the same original size bins and 31 blocks: 6964 baseline objects, of which 4964 have positive finite A1. A binomial model of endpoint observability includes treatment, large-size status, their interaction, log baseline area, baseline detector score and donor fixed effects, with donor-cluster uncertainty. The interaction coefficient is -0.4479 (odds ratio 0.639, 95% cluster interval -0.788 to -0.108 on logit scale). Donor-equal medians of block-level treatment minus DMSO observability gaps are +0.052 for small and -0.052 for large, with large-minus-small gap -0.093. A baseline high-detector-score negative-control outcome has interaction odds ratio 0.976, cluster $p=0.805$; it does not remove unmeasured batch confounding.

The treatment-versus-DMSO observability gap is relatively more favorable for small than large baseline objects in this conditional table, after accounting for specified covariates. This is a detection/retention result, not a directionally identified bias in swelling: the unknown A1 of absent endpoints can be high or low, and never-detected baseline objects are outside the frame. Donor and experiment effects were collinear in the restricted original-block design, so the final model uses donor fixed effects without experiment effects; only 12 donor clusters make model p-values descriptive. Per-block four-cell rates, per-donor medians, source hash and controls are in results/endpoint_observability.json and scripts/audit_endpoint_observability.py. Neither this model nor the previous hypothetical curve establishes biological invariance, independent replication or a changed sign gate.

4.3l Missing-endpoint percentile envelope

The prior one-arm tipping scenario probes one missingness pattern; it does not bracket all possible missing outcomes. In a separate post-result partial-identification exercise, we assigned each missing A1 log response a value bounded by the pooled observed 5th and 95th percentiles in its treatment-by-size cell. Choosing the adverse endpoint separately in each of four cells gives an equal-donor mean of donor-median block contrasts as low as -0.745, with 0/12 positive donors; reversing every bound gives as high as +1.362, with

12/12 positive. Zero is inside this wide conditional envelope. By contrast, the observed-positive-pair mean is +0.208 and derived-only mean +0.195. Even if unobserved responses were restricted to the broad observed central 90% of their own cells, the sign of the size-treatment contrast would not be identified by these source tables.

The 5th-95th percentile assumption is hypothetical and chosen after seeing the source distribution. It is not proof that absent objects have outcomes inside those ranges; indeed, segmentation failures may select extremes. The calculations preserve all observed positive A1 outcomes and change only the missing entries in the 31 original blocks. Original masks and objects never detected at baseline remain unavailable. This honest envelope is not a new negative experimental result, a p-value or a license to rewrite the original biological gate. Full per-donor interval endpoints, source hash, protocol and script are in `results/missing_endpoint_bounds.json`, `notes/postresult_missing_endpoint_bounds.md` and `scripts/bound_missing_endpoint_response.py`.

4.3m Well-balanced source-table check

Source-table organoids from the same well are nested technical replicates. To test whether high-count wells dominate the observed-endpoint contrast, we fixed this post-result comparison before running it (`notes/postresult_well_balance_protocol.md`): take every positive-A0/positive-A1 source pair in the 31 original blocks and original size bins, compute each well's mean $\log(A1/A0)$, then weight observed wells equally within each arm-by-size cell. The original object-weighted mean is recovered at +0.207978 for the equal-donor mean of median block effects, with 9/12 positive donors. Equal-well weighting gives +0.196085 with 9/12 positive donors; its difference from object weighting is -0.011893 log units. All 31 blocks retain four observed arm/size cells. The predeclared ≥ 0.10 change, donor sign-count change, or missing-cell sensitivity flag was not triggered. This only shows little influence from unequal counts among wells with observed endpoints, not robust biological treatment-by-size response.

We separately retained all 6,964 baseline-detected objects, including the 2,000 without a positive endpoint, to calculate the treatment-by-size interaction in endpoint-observability rates. Object-weighting gives -0.0932 and equal-well weighting -0.0645, both with 4/12 positive donor medians. Well-balanced observability does not identify the missing A1 values. Well counts vary from one to three in individual cells, and well grouping does not give independent donor replication. Per-cell counts, source hashes and executable code are in `results/well_balance_sensitivity.json` and `scripts/audit_well_balance.py`. This same-accession diagnostic cannot revise the failed original 9/12 sign gate or no-forskolin specificity gate; the prior missing-endpoint envelope still includes zero.

4.3n Baseline image-edge proxy and endpoint visibility

The source's baseline detection boxes permit a narrow test of whether proximity to the 2048-pixel image boundary predicts an absent endpoint. We locked a box-edge clearance below 50 pixels and a fixed 100-pixel sensitivity before calculating the association (notes/postresult_edge_observability_protocol.md). Among 6,964 baseline-detected organoids in the original size/arm cells, 404 have a box within 50 pixels of an edge. At 50 pixels, 24 of the 31 original blocks contain both edge and interior objects in all four arm/size cells, spanning 11 donors; the other seven blocks are listed without substitution in results/edge_observability.json. Thus the block-balanced edge comparison is underpowered and incomplete by construction.

A binomial endpoint-observability model with baseline area, detector score, donor factors, treatment, size, treatment-by-size, edge, and edge-by-size/treatment terms converged. The edge main-effect odds ratio is 1.136 (donor-cluster interval on logit scale -0.171 to +0.425); edge-by-size odds ratio 0.813 and edge-by-treatment odds ratio 0.868. Their wide intervals cover zero on logit scale. The treatment-by-size endpoint-observability odds ratio stays 0.638, versus 0.638 under the fixed 100-pixel proxy. This is not evidence that boundary cropping is absent: box clearance does not show whether pixels or masks were cropped, and only 404/6,964 objects meet the 50-pixel indicator. Baseline-undetected organoids, absent endpoint areas and real image artifacts remain unmeasured. Full four-cell rates, excluded-block identities, hashed source and code are in results/edge_observability.json and scripts/audit_edge_observability.py. No original biological gate changes.

4.3o Paired-center displacement and track-selection support

The source table stores paired x/y centers only when both segmented areas are positive. Before computing this post-result influence check, we fixed displacement bins [0,10), [10,25), [25,40), and [40,50] pixels, then a <40-pixel subset (notes/postresult_displacement_sensitivity.md). Across all 29,648 positive-area pairs under the released source, the maximum displacement is 50.0 pixels and none exceed 50. This sharp support boundary suggests a pairing restriction, but source tracking code has not been established and the limit cannot be called its documented algorithm. For the 4,964 source-positive pairs in the original 31 size/arm blocks, the all-pair equal-donor mean of median block contrasts is +0.207978 (9/12 positive donors). Restricting observed pairs to displacement <40 yields +0.212476 with 9/12 positive donors, but one fixed block loses the minimum three observations in one cell. Comparing exactly the same 30 retained blocks, all pairs give +0.213675 and the <40 subset differs by -0.001199. This is not a newly chosen eligibility gate or an independent replication.

Within the DMSO-only observed pairs, the pooled displacement-response Spearman correlations are +0.069 for small and -0.060 for large; equal-donor means of within-well median correlations are -0.050 and -0.047. The paired-center boundary is a reproducible

property of these selected rows, not proof of correct track identity or absence of selection bias. Without x/y for 2,000 baseline-detected objects missing positive A1, no displacement analysis can recover their swelling responses or test whether long-moving organoids were excluded. The full four-cell displacement histograms, lost-block ID, source hash and script are in results/displacement_sensitivity.json and scripts/audit_displacement_sensitivity.py. This audit leaves the original failed donor sign and zero-forskolin gates untouched and cannot tighten the hypothetical missing-response envelope.

4.3p Baseline neighbor spacing and endpoint observability

A post-result, protocol-locked source-table audit measured the distance from each baseline-detected organoid to its nearest OTHER baseline-detected object in the same image, including middle-size objects in the neighbor pool (notes/postresult_neighbor_observability_protocol.md). The analysis frame is the original 31 positive-dose blocks and extreme-size cells: 6,964 baseline objects in 143 source image paths, with 4,964 positive finite endpoints. Nearest-distance quartiles are 70.9, 104.2, and 152.1 pixels. A descriptive donor-fixed-effects binomial model adjusted for arm, size, their interaction, baseline log area and detector score finds an odds ratio 1.724 for a unit increase in $\log(1 + \text{nearest distance})$ in small objects (12-donor cluster interval on the coefficient +0.414 to +0.676); the size-by-distance ratio of odds ratios is 1.063, with an interval crossing 1. These are conditional object-level associations, not independent biological replicates.

Within 143 image paths, the closer half's mean positive-endpoint rate minus the farther half's rate is -0.101; 100 seeded within-field permutations of proximity labels give a descriptive 95% technical-null range [-0.018, +0.021]. Thus source baseline spacing marks an observability gradient, not a proven causal crowding or tracking mechanism. The source has paired $t=0/t=1$ keys even when A1 is blank; 2,000 endpoint responses remain unobserved, baseline-undetected organoids are outside the frame, and field paths may represent repeated acquisitions. Neither the failed original donor sign and zero-forskolin specificity gates nor the zero-crossing missing-response envelope changes. Full counts, source hash and code are in results/postresult_neighbor_observability.json and scripts/audit_postresult_neighbor_observability.py.

A retrospective composition check held image path, treatment arm and original size bin fixed before dividing each eligible stratum at its own nearest-neighbor median (notes/postresult_neighbor_stratified_sensitivity.md). 279 strata cover 6,952 of 6,964 frame objects; 5 sparse strata with 12 objects are reported, not silently dropped. The closer-minus-farther endpoint-observability difference is -0.082 with strata weighted equally, -0.068 when weighted by stratum object count and -0.077 when donor means are weighted equally (12/12 donor means negative). The largest cell-level gradient is the treated-small cell at -0.172; all four cells' counts are recorded. One hundred seeded

within-stratum proximity-label permutations yield a technical-null 95% range [-0.023, +0.024] for the equal-stratum mean. The gradient therefore is not solely the pooled arm/size composition in this source frame. It is still a retrospective association, not a treatment-effect correction, tracking-causation test, or an answer for the 2,000 absent positive endpoints.

4.3q What the observed tracks identify, and what they do not

The selection audit has three distinct denominators. The released paired-particle table contains baseline-detected organoids, including those with a blank or nonpositive endpoint; the derived response table contains a narrower subset with usable A0 and A1; the actual well may also contain organoids never detected at baseline, which neither table enumerates. These populations cannot be treated as interchangeable. The following accounting table uses only the original 31 donor-by-experiment-by-dose blocks and four pre-existing treatment/size cells, not a newly selected donor population. Its observed-endpoint fractions condition on baseline detection; they are not treatment-response rates.

Table 11. Source-frame denominator audit for the four original cells. Percentiles describe selected positive-endpoint tracks and are assumptions, not known support bounds, for absent responses. Source: results/missing_endpoint_bounds.json and tabular_track_retention.json.

arm	baseline size	baseline detected	positive A1	missing A1	visible fraction	observed log-response 5-95%
DMSO	small	1248	563	685	0.451	[-0.351, +1.157]
DMSO	large	2058	1825	233	0.887	[-0.415, +0.992]
Trikafta	small	1250	571	679	0.457	[-0.258, +1.415]
Trikafta	large	2408	2005	403	0.833	[-0.005, +1.557]

The difference between small and large organoids is not a minor rounding issue. Conditional positive-endpoint visibility is about 45% in small DMSO and 46% in small treated objects, versus 89% and 83% in the respective large cells. The treatment gap in visibility reverses across these size strata; the donor-equal gap contrast is -0.093. A donor-adjusted observability model that also includes log baseline area and detector score estimates a treatment-by-large odds ratio 0.639. The baseline high-detector-score control has the analogous odds ratio 0.976. These are features of a selected imaging pipeline, not evidence that missing small objects would swell more or less. Donor fixed effects and twelve clusters do not remove plate, imaging, field, or track-assignment confounding.

The original biological estimand is the donor-level drug-minus-DMSO contrast in large versus small log swelling. It differs from the estimand of the observability model, which is whether a baseline-detected object receives a positive A1. Conditioning the swelling analysis on a positive A1 may induce selection dependence between initial size and an unmeasured cause of endpoint detection. An inverse-inclusion model fitted among rows with both positive areas can address only selection into the derived table among those already observed; it cannot recover the 2,000 absent positive endpoints. This is why the small shift in cross-fit inclusion-weighted means is not a missing-data correction, and why a baseline score control does not identify unmeasured endpoint values.

A separate spacing audit gives a measured example of this selection structure. Within image-path, treatment and size strata, closer baseline neighbors have an equal-stratum positive-endpoint rate 0.082 below farther neighbors; 279 eligible strata retain 6,952 of 6,964 source-frame objects and all twelve donor-equal differences have the same negative direction. Distance is a baseline geometry proxy, not a randomized crowding intervention: object density, image quality, local tracking competition and acquisition repetition can all generate it. A field-stratified association is informative about where data are lost, but does not estimate the missing swelling response or the biological size mechanism.

4.3r Conditional missing-response bounds across donors

To expose how much the missing responses matter at the donor level rather than only in a pooled interval, The following table lists the fixed-donor contrast under three previously committed analyses. The first uses only source pairs with positive A0 and A1; the latter two impute each missing endpoint at an adverse or favorable observed cell-specific percentile. These endpoints are conditional scenario bounds, not formal identification bounds: the 5th and 95th percentiles of observed tracks need not contain the missing outcomes. The same donor, experiment, dose and size eligibility from the original blocked test is kept throughout.

Table 12. Fixed donor-level median block contrasts in log swelling. Scenario columns do not constitute confidence intervals or imputed outcomes; source: results/tabular_track_retention.json and results/missing_endpoint_bounds.json.

donor	observed positive-pair contrast	adverse percentile scenario	favorable percentile scenario
PDIO_01	+0.039	-0.669	+1.304
PDIO_03	+0.351	-0.379	+1.150
PDIO_05	+0.540	-0.468	+1.341
PDIO_06	+0.371	-0.619	+1.540
PDIO_09	+0.201	-0.449	+1.281
PDIO_11	+0.366	-0.455	+1.258
PDIO_12	+0.418	-0.667	+1.351

PDIO_13	+0.273	-0.944	+1.685
PDIO_14	-0.193	-0.969	+1.434
PDIO_15	+0.301	-1.025	+1.369
PDIO_16	-0.097	-1.328	+1.391
PDIO_18	-0.073	-0.967	+1.235

Across donors, the equal-donor mean of observed positive-pair contrasts is +0.208; the adverse and favorable percentile scenarios span -0.745 to +1.362. No donor remains positive in the adverse scenario; all twelve are positive under the favorable one. Because zero lies in this conditional envelope, source-table endpoint observability alone cannot establish the direction of the full baseline-detected-population effect. These scenarios alter only the missing A1 values; they cannot address organoids absent at A0, incorrect track links, size-dependent area error or the failed zero-forskolin specificity test.

A less adversarial one-arm tipping exercise holds three missing cells at their observed medians and increases only the hypothetical small-treated missing log response. The equal-donor mean reaches zero after a +0.61 log-unit shift; this is a scenario threshold, not a fitted estimate of those missing responses. Its direction matters: greater swelling among missing treated-small objects raises the small-treatment effect and can erase the observed large-minus-small difference. The percentile envelope is broader because it permits all four cells to vary together. Neither calculation selects a correction to apply to clinical data.

The next discriminating measurement is not another threshold tuned to this accession. A new acquisition should enumerate all baseline objects before filtering, retain raw masks at both time points, identify fields/plates/donors, audit possible one-to-many tracks, and include positive, zero-forskolin and CFTR-inhibitor controls. Lock the size function, donor-level endpoint, missingness model and analysis plan before new outcomes are opened. Until then, the substantive result of the present pipeline is a quantified selection problem and a within-accession size association with failed companion gates, not a verified CFTR-specific size mechanism or a patient predictor.

4.4 Does size correction help theratyping? No

Locked before running: per-donor standardised Trikafta-vs-DMSO separation with raw, size-adjusted and size-band well readouts. Median raw 4.73, adjusted 4.35, band 4.39; adjusted better in 8 of 17 donors. Excluding organoids below 1,069 px (a post-hoc threshold): median 4.73 -> 4.75, better in 9 of 17, Wilcoxon $p = 0.68$. The size trend does not improve calls, and the stricter matched-block H1 fails.

Secondary exploratory analyses II: segmentation benchmark and split integrity

Sections 4.5 through 4.5j are a secondary arm. Its benchmark-beat claim is withdrawn because repeated acquisition fields cross the splits; the arm is retained as an integrity case study and labeled exploratory.

4.5 Segmentation benchmark

Table 13. Instance segmentation on the OrgaSegment eval split (12 images).

model	eval mAP@0.5	sd	source file
U-Net v1, 256 px, default post-processing	0.634	0.157	results/seg_eval.json
U-Net v1 + val-tuned post-processing + flip TTA	0.710	0.127	results/seg_eval_tuned_256.json
U-Net v2, 512 px, 60 epochs, default post-processing	0.691	0.135	results/seg_eval_512.json
U-Net v2 + val-tuned post-processing + flip TTA	0.733	0.129	results/seg_eval_tuned_512.json
OrgaSegment (published, Mask R-CNN)	0.760	0.120	Lefferts et al. 2024

Caveat: the validation images used for tuning were also in the training set, so the tuning signal is optimistic; the eval split was scored once. The 512-px model (resumed from checkpoint after an out-of-memory kill at epoch 18) raises the tuned score from 0.710 to 0.733, still below the published mean. With 12 eval images and sd near 0.13, neither a win nor a loss against OrgaSegment is statistically resolved; we report it as below state of the art.

4.5a Segmentation split-integrity correction

Training combined 184 train and 35 validation images. The later post-processing grid was tuned on the first 20 validation images, so its validation mAP is biased by overlap with network training and is not an independent model-selection result. The filename-distinct 12-image eval split was scored separately, and its mAP@0.5 remains 0.733 versus published 0.76. Bootstrap across our eval images gives CI [0.663, 0.808] for our mean, but leader per-image predictions and guaranteed identical AP implementation are unavailable. No benchmark break (results/seg_integrity_audit.json).

4.5b First clean train-only and val-only segmentation attempt: negative

A separate protocol was frozen before fitting (results/preregistration_trainonly_seg.md): 184 training images only, 35 filename-distinct validation images, 12 evaluation files not scored until final selection, but repeated acquisition fields were later found across splits (Section 4.5e). Sixty epochs were fit; epoch 40 minimized validation cross-entropy. A fixed grid of 108 postprocessing choices used the first 20 validation images only,

selecting [0.4, 0.5, 0.3, 40] at validation AP50 0.811. This validation score is a selection statistic, not an independent benchmark.

The single sealed final evaluation scored 0.73867 AP50 under the authors' released Cellpose-style scorer and 0.73933 under our unique-mask-count scorer, below the published 0.76. The only material matcher difference arises from one released GT mask with a skipped label ID: the original scorer uses the maximum ID as object count. This small numerical difference is not a model gain. Author baseline per-image predictions were not run in this 2-CPU, no-TensorFlow/GPU workspace, so we do not assert a paired statistical comparison. Clean selection fixes an older exact-file train/val overlap, but the later near-duplicate audit finds that it did not secure acquisition-group independence. It is not a benchmark match. Per-image TP/FP/FN and both AP versions are preserved in results/trainonly_seg/sealed_eval.json; selection frozen before eval at commit 120f944.

Table 14. Sealed 12-image evaluation of the clean train-only model; image numbers follow the JSON per_image order, which has full filenames and matching counts. None selected the model or postprocessing.

image #	author AP50	unique-count AP50	GT masks	predicted masks
1	0.958	0.958	70	69
2	0.585	0.585	76	54
3	0.796	0.796	44	53
4	0.892	0.892	36	34
5	0.836	0.836	52	60
6	0.585	0.585	50	53
7	0.709	0.709	149	145
8	0.649	0.649	128	93
9	0.685	0.685	75	80
10	0.737	0.737	59	73
11	0.652	0.652	148	113
12	0.780	0.788	86	91

4.5c Locked v2 train-only improvement and narrow published numerical threshold

The negative clean v1 checkpoint above remains the historical result, not a deleted failure. Amendment-free preregistration in notes/prereg_ap50_beat.md locked the v2 gate at mean author-scorer AP50 > 0.76 on the same nominally held-out 12-file eval split. A later scene-overlap audit invalidates its independence (Section 4.5e). Fine-tuning trained solely on 184 train images for 30 extra epochs with stronger augmentation and a lower learning rate; the checkpoint at epoch 30 minimized cross-entropy on the filename-distinct but scene-overlapping validation split. The unchanged 108-combination grid was scored only on the first 20 validation images. Grid index 41 selected [0.5, 0.5, 0.3, 40] at

validation AP50 0.826061; selection was frozen at commit 7eaa46c17a610ac8173ad1194d992668403a6760 before one v2 evaluation-file score.

The released author scorer (OrgaSegment v1.0.1 commit aeaec17c42fefde61a8b2215a616d419e8126e9f) gives an evaluation-file mean AP50 0.761108, exceeding the locked numerical comparison 0.760000 by 0.001108. The unique-mask-count implementation gives 0.761773 and is NOT used for the gate. These are the saved single-run per-image means in results/trainonly_seg/sealed_eval_v2.json, not replicate-run means or a confidence-bound comparison. The released Mask R-CNN predictions were not scored head-to-head here; the published rounded reference 0.76 +/- 0.12 may be an estimate from a differing evaluation protocol. The numerical margin is small relative to image-to-image spread. More importantly, repeated fields of view span train and eval files (Section 4.5e): the >0.76 numerical crossing is preserved as a historical calculation but cannot count as an independent benchmark win. The model cannot be called the best virtual organoid in the world.

Table 15. V2 single sealed author-scorer evaluation. Full image names, TP, FP, FN, and mask-ID-gap accounting: results/trainonly_seg/sealed_eval_v2.json. This is a frozen-selection evaluation-file comparison, but acquisition-field overlap disqualifies an independent held-out interpretation.

image #	author AP50	unique-count AP50	GT masks	predicted masks
1	0.958	0.958	70	69
2	0.658	0.658	76	55
3	0.796	0.796	44	53
4	0.944	0.944	36	34
5	0.879	0.879	52	57
6	0.667	0.667	50	50
7	0.729	0.729	149	145
8	0.630	0.630	128	92
9	0.707	0.707	75	82
10	0.711	0.711	59	71
11	0.665	0.665	148	110
12	0.790	0.798	86	92

Open controls: release the prediction masks and rerun both models under exactly identical code and split; estimate paired uncertainty over images and train seeds; test a separately sourced annotation cohort. No threshold, checkpoint, or postprocessing retuning after the sealed v2 result is permitted for a claim about this same eval split.

4.5d What the thin numerical threshold crossing cannot establish

After the sealed v2 result, we wrote and ran a descriptive uncertainty audit; it is explicitly post-evaluation and cannot change the predeclared >0.76 numerical gate. The 12-image

author-scorer mean is 0.761108 with sample image SD 0.113212. A seed-20260926 image-resampling percentile interval from 20,000 draws is [0.703230, 0.825010]. It includes the published rounded 0.76; it does not account for acquisition-field leakage, and 5/12 leave-one-image-out v2 means fall at or below 0.76. All image deletion outcomes are preserved in results/ap50_uncertainty_audit.json; none are used to reselect a model. This is evidence of thin numerical margin; the cross-split repeated fields independently disqualify a held-out general-superiority claim.

The paired v2 minus our own clean v1 image mean is +0.022436, with conditional image-bootstrap interval [+0.004474, +0.041408]. It improves 8 images, declines 2 and ties 2 under a $1e-7$ tolerance; the two-sided sign p over nonties is 0.109375. This within-project comparison is useful debugging evidence, not a head-to-head result against the released Mask R-CNN or a second cohort. Image bootstrap estimates uncertainty conditional on this fixed split and ignores training-seed and domain-shift variation. A new acquisition batch with locked weights/preprocessing and, ideally, paired author checkpoint predictions is the needed confirmatory benchmark.

4.5e Post-result audit: repeated acquisition fields cross segmentation splits

A new post-result forensic scan found zero identical image or mask SHA256s across the 184/35/12 train/val/eval files. But standardized 32-by-32 grayscale thumbnails flagged 5 cross-split pairs above cosine 0.95. The top train/eval pair reached 0.996584. Direct visual inspection of the original image pixels confirmed the same organoid layout in this pair and in two train/val pairs despite different filenames, brightness and image detail. A fourth val/eval pair has a partly shifted but similar field; its acquisition identity is not adjudicated. This scan is post hoc and the threshold is an investigation aid, not a validated group identifier. The complete pair list and file hashes are in results/seg_split_integrity_v2.json; screenshots are not needed to reproduce the thumbnail audit from the original public files.

This changes the interpretation of Sections 4.5a-d. Train-only means train-filename-only, not acquisition-scene-only. File-untouched means scores were not requested during selection, not that the underlying scenes were unseen. The 0.761108 author-scorer mean and the locked >0.76 numerical arithmetic remain exactly as recorded; the prerequisite of an independently held-out benchmark sample was not satisfied. We retract the benchmark-improvement claim rather than excluding overlapping eval images after seeing outcomes, rewriting the gate, or treating a post hoc subgroup as a new sealed test. The earlier 0.73867 negative remains in the record. We need source acquisition and donor metadata, group-disjoint training/validation, an untouched independently acquired evaluation cohort, and a paired author-baseline scorer before claiming benchmark improvement. Source-derived filenames are not enough to prove independence of remaining images.

4.5f A scene-graph audit and its false positives

We implemented an image-only graph for all 231 images, with every pair's 32-by-32 standardized thumbnail cosine and an exact-hash check. Connected components are candidate acquisition scenes; their transitive closure retains repeated frames. At threshold 0.95 there are 215 components, 14 multi-image components, and 5 cross-split components comprising 10 images. At 0.97 and 0.99, 4 and 3 cross-split pairs persist. All cutoffs were selected or inspected AFTER the saved AP50 outcome and so are forensics, not preregistered evaluation choices (scripts/build_scene_graph.py, results/scene_graph_forensics.json).

A seeded random sample of 1000 cross-split pairs has median cosine 0.352 and 99th percentile 0.865; this descriptive scale check does not yield a calibrated p-value. Visual inspection of the five flagged pairs at 0.95 confirmed three near-identical organoid layouts (one train/eval and two train/val). One val/eval pair may be a shifted scene and remains uncertain. The fifth, a 0003 train/val pair near 0.95065, shows visibly distinct layouts: it is likely a similarity false positive. These adjudications were based on the pixels, not model outcomes; the graph retains even the false-positive candidate to expose threshold limitations. No missing graph edge proves different donors, plates, wells or sessions, because that metadata is unavailable. The old test files have already been seen; excluding image pairs now cannot create a sealed holdout. A new group-disjoint acquisition cohort and published-model paired predictions are still required.

4.5g Repeated-field labels confirm shared organoids

The three strongest cross-split image pairs, all detected above 0.99 cosine before opening their masks in this follow-up, also share labeled organoids. We matched raw 512-by-512 instance masks one-to-one at identical pixel coordinates with a maximum-IoU assignment. Respectively, the two train/val pairs contain 61 and 40 matches at $\text{IoU} \geq 0.5$ out of 62/66 and 43/45 annotated objects; the train/eval pair contains 50 matches out of 55/52 objects. Corresponding foreground Dice values are 0.905, 0.809, and 0.896. This is direct evidence that many identical organoid layouts and corresponding mask annotations cross filenames and splits, not merely similar illumination or texture.

The visually distinct-layout 0003 train/val candidate at cosine 0.95065 yields 0 matched objects and Dice 0.028. A retrospectively selected 512-pixel train/eval distinct-layout comparison also yields 0 matches; controls illustrate that thumbnail cosine alone makes false-positive candidate scene groups. The mask checks were performed only after the published-file AP50 was scored; they are annotation consistency evidence, NOT a prospective segmentation test or estimate of benchmark inflation. The same source masks may have been curated together and cannot establish donor independence, either. All original pairs and descriptive controls are in scripts/audit_scene_label_consistency.py and results/scene_label_consistency.json. We have not removed overlapped images or rescored a post hoc uncontaminated subset.

4.5h Matched organoid area and boundary variability across repeated fields

The three high-similarity cross-split fields also permit a direct comparison of their separately stored instance boundaries and areas. A fixed post-result descriptive protocol was written for this analysis; it is not an independently held-out test (notes/paired_scene_boundary_protocol.md).

At identical raw 512-pixel coordinates we matched labels one-to-one by $\text{IoU} \geq 0.5$. There are 61, 40, and 50 matched organoids in the three previously flagged pairs. Their median absolute log area ratios are 0.1086, 0.2934, and 0.1451; the median one-pixel-tolerant boundary F1 values are 0.7207, 0.4694, and 0.7314. An exact-copy metric control returns zero area/centroid drift and boundary F1 1; two visually distinct-layout pair controls have zero matched objects at $\text{IoU} \geq 0.5$. The code and every per-object measurement are in scripts/audit_paired_scene_boundaries.py and results/paired_scene_boundary_consistency.json. This is a measurable repeated-field annotation/acquisition variability signal, not a human inter-rater experiment: acquisition time, annotator identity, mask-curation history and donor IDs are unknown. No valid AP50 ceiling or CFTR phenotype-robustness result follows from it. Independent image acquisition and blinded repeat annotation would be required for those claims.

4.5i Retrospective overlap-exclusion score audit

A further post-result audit asks the narrowest question still available from already-saved outputs: if the graph-flagged cross-split eval images are removed after all scores were viewed, what happens to the saved author-scorer mean? Exclusions come only from the existing image-only graph, never from outcome values. At thresholds 0.95 and 0.97 the flagged eval images are the train-overlapping 20210727_11562 file (AP50 0.87931) and the val-overlapping 20210727_26838 file (AP50 0.71053); removing them leaves 10 images with mean 0.754346, versus 0.761108 on all 12 and 0.729115 for the earlier v1 model on the same retained images. At 0.99 only the train-overlapping file is flagged, leaving 11 images at mean 0.750362. Seeded image-bootstrap intervals, [0.690968, 0.827025] and [0.692235, 0.816450], stay wide and include the rounded published 0.76. Protocol, input hashes and every retained and flagged score are in notes/postresult_scene_overlap_score_audit.md, scripts/audit_postresult_scene_overlap_score.py and results/postresult_scene_overlap_score_audit.json.

None of these subset means is a held-out benchmark score: the eval files, model and grid were all fixed and scored before the exclusion rule existed, unflagged files can still share acquisition sessions or donors, and no published author-model predictions exist for a paired comparison. The audit is reported because it removes a tempting misreading: even the most aggressive outcome-aware exclusion does not rescue the comparison, it lowers the retained mean below the rounded 0.76 point. The locked all-image v2 score stays

exactly as recorded, the retained-subset numbers are diagnostics, and the requirement for a new group-disjoint acquisition cohort with paired author-baseline scoring is unchanged.

4.5j Distinct mouse-intestinal detection transfer: negative

We tested the fixed human-organoid segmentation checkpoint on a different image and annotation domain, not as a repaired version of the original human mask benchmark. The Domènech-Moreno mouse-intestinal 4x EVOS release (Zenodo 6768583) contains 756 train and 84 released validation JPEGs with manually drawn YOLO object boxes across four morphology classes. Its 193.9 MB archive matches the source MD5. Before reading target scores we fixed the existing v2 U-Net weights and the original human-validation parameters, converted its instance masks to tight object boxes, collapsed the target classes into one object class and matched boxes once at $\text{IoU} \geq 0.5$. Across all 84 validation images and 2,468 valid labeled boxes, there are 850 true positives, 911 false positives and 1,618 false negatives. Micro precision 0.483, recall 0.344, F1 0.402; mean per-image F1 0.340. One released zero-width annotation was recorded and excluded from the valid-box denominator. The source hashes, every image count, protocol and implementation are in `results/postresult_mouse_yolo_domain.json`, `notes/postresult_mouse_yolo_domain_protocol.md` and `scripts/audit_mouse_yolo_domain.py`.

This is an external-domain negative for the current model's ready-to-use object detection. The target is mouse rather than human, class-specific loose boxes rather than pixel instance masks, and a different microscope/field density, so box F1 cannot be compared with the OrgaSegment paper's human mask AP50 0.76. No published comparator's predictions on the same 84 images are in hand, so this is NOT a leading-tool benchmark. No exact image-hash duplicates occur across target train/validation or original human source, but thumbnail similarity flags possible related target acquisition fields and lacks donor/session metadata. Because no mouse images trained our U-Net, this issue does not explain the low transfer recall, but it prevents an independent mouse split claim. Visual review of example low- and higher-scoring target images showed substantial contrast and object-density differences without identifying the cause of false negatives. The result narrows tool deployment; it neither tests CFTR swelling biology nor alters the failed original donor sign and specificity gates.

4.5k Candidate-scene-group-disjoint reassignment and its exploratory test

We made the image-based leakage diagnosis operational, while preserving the original result as history. Under the 27 September exploratory protocol (`notes/prereg_group_disjoint_seg.md`, commit b8e5132), an exact image hash or standardized 32-by-32 thumbnail cosine ≥ 0.95 joined images into candidate scene groups, including transitive links. Each group's reassignment followed the highest original priority $\text{eval} > \text{val} > \text{train}$. Five files moved, including one train file to eval and one val file to eval. The resulting 180/37/14 train/val/eval allocation includes all 231

released files exactly once and puts no graph component across splits (results/group_disjoint_seg_split.json, split implementation commit db30071). At cosine 0.97 and 0.99, the earlier forensic graph found 4 and 3 originally cross-split candidate groups, versus 5 at 0.95; these are descriptive threshold checks, not alternative assignments chosen by AP50. The 0.95 graph is a proxy: one flagged pair has visually different layouts and unflagged files can still share donors, wells or sessions. Without acquisition metadata, this reassignment cannot certify independence.

We freshly initialized the 3-class base-16 U-Net and trained on only the 180 reassigned train files, with no weights from the old v2 checkpoint. The frozen recipe used 512-pixel grayscale normalization, 256-pixel random crops, batch size two, weighted cross entropy 1/1/3, Adam at 0.002, seeded flips and rotations, and up to 60 epochs. The 37 reassigned validation files selected epoch 60 by lowest weighted cross entropy checked every five epochs (loss 0.077038895); their masks did not enter gradient updates. The checkpoint SHA256 is cc9f8ddfd07155854c8955ccd3089de777bd53fcf6131d0ef7aadb26142360f4. The old postprocessing grid was not rerun: script commit 6b23675 froze the code-default foreground/interior/boundary/minimum-size parameters [0.5, 0.6, 0.3, 20] before the first reassigned-eval score, as permitted by that protocol. Model checkpoint and full epoch history are retained under results/group_disjoint_seg/.

On the 14 reassigned eval images, the pinned OrgaSegment v1.0.1 author AP50 implementation gives mean 0.750403; our unique-mask-count implementation gives 0.751727. A seed-20260927 image bootstrap with 20,000 draws produces a conditional percentile interval [0.689349, 0.810722]. This includes the rounded published 0.76 and omits seed and donor/acquisition variance. The two scorers differ beyond float rounding on two images with missing ground-truth label IDs: the author scorer counts maximum instance label, while ours counts distinct labels. Neither metric is a head-to-head comparison with released Mask R-CNN predictions, which are unavailable here.

Table 16. Candidate-group-disjoint exploratory reassignment: all 14 image scores, including the two moved-to-eval files. Full filenames, unique-count scores and mask-ID-gap accounting are in results/group_disjoint_seg/exploratory_eval.json.

eval image #	author AP50	TP	FP	FN
1	0.958	68	1	2
2	0.583	49	8	27
3	0.811	43	9	1
4	0.892	33	1	3
5	0.895	51	5	1
6	0.554	36	15	14
7	0.724	126	25	23
8	0.850	51	5	4
9	0.632	86	8	42

10	0.702	66	19	9
11	0.767	56	14	3
12	0.705	55	19	4
13	0.652	103	10	45
14	0.780	78	13	9

Failure analysis is heterogeneous rather than a single global threshold effect. The lowest-scoring images had author AP50 0.554 (36 TP, 15 FP, 14 FN) and 0.583 (49 TP, 8 FP, 27 FN); the highest reached 0.958 (68 TP, 1 FP, 2 FN). The two moved-to-eval images scored 0.850 and 0.767, but they were already available to prior analyses and cannot be called unseen. On the 12 original eval images shared with the old v2 report, a descriptive same-file paired mean change is -0.0204 AP50. This is not an isolated effect of leakage removal: weights, training membership, validation set, model selection and extraction parameters all changed. The original 0.761108 result and numerical threshold crossing remain archived; the new 0.750403 estimate is not a corrected benchmark score, an independent replication or proof that the old estimate was inflated by exactly their difference. The eval includes 10 singleton proxy groups and two two-image groups, but these are not verified independent donors or plates; the image-bootstrap interval ignores any hidden nesting. An untouched acquisition cohort with real donor/plate/field IDs and paired published-model predictions is still required for a leading-tool claim (notes/group_disjoint_eval_design_limits.md).

Secondary exploratory analyses III: organoid-to-tissue fidelity across public transcriptomes

Sections 4.6 through 4.13 are off-question secondary analyses of organoid-to-tissue fidelity in public expression data, retained and labeled exploratory.

4.6 Organoid-to-tissue fidelity across 151 GEO series

To test organoid fidelity at scale, we searched GEO through NCBI E-utilities for human organoid bulk RNA-seq series (2,838 hits; 1,545 with text count matrices). For 151 series with a usable gene-level matrix and a title naming a single organ, each sample was converted to log counts per million and averaged. The series profile was then correlated with every GTEx v8 tissue median profile (54 tissues).

$$CPM_{gs} = 10^6 x_{gs} / \sum_g x_{gs}, \quad p_g = (1/S) \sum_s \log(1 + CPM_{gs}) \quad (11)$$

$$r_{t^{cent}} = \frac{\langle p - \mu_p - (m - \mu_m), G_t - m - \text{mean}(G_t - m) \rangle}{(\| \cdot \| \cdot \| \cdot \|)}, \quad m_g = \text{mean}_t G_{gt} \quad (12)$$

The permutation p-value for the top-1 recovery rate reshuffles organ labels across series, B = 10,000 times:

$$p_{perm} = (1 + \#\{b : \text{top1}_b \geq \text{top1}_{obs}\}) / (1 + B) \quad (13)$$

Uncertainty on shares uses a series-level percentile bootstrap:

$$CI_{.95} = [q_{.025}, q_{.975}] \text{ of } \{f(D^*_b)\}, D^*_b \text{ drawn with replacement from the series set, } b = 1..2000 \quad (14)$$

Table 17. Tissue recovery and fibroblast best-match share. S3000: Spearman on the 3,000 most tissue-variable genes; SALL: all genes; CENT: GTEx-centred Pearson; _NOCULT: 853 MSigDB Hallmark proliferation/MYC/EMT genes removed. Clean subset: 114 series whose summary confirms a single organ and mentions organoids (113 with profiles). Files: results/geo_fidelity_methods.json, results/geo_fidelity_clean.json.

method	top-1 (all)	null top-1	p_perm	top-1 (clean)	fibroblast best (clean)	95% CI
S3000	0.272	0.060	1e-04	0.301	0.230	[0.15, 0.31]
SALL	0.331	0.080	1e-04	0.327	0.257	[0.18, 0.35]
CENT	0.252	0.056	1e-04	0.274	0.363	[0.27, 0.45]
SALL_NOCULT	0.331	0.079	1e-04	0.327	0.248	[0.17, 0.33]
CENT_NOCULT	0.258	0.059	1e-04	0.274	0.425	[0.34, 0.51]
SALL_NONPAR	0.417	0.103	1e-04	0.398	0.204	[0.13, 0.28]
CENT_NONPAR	0.331	0.088	1e-04	0.345	0.372	[0.28, 0.46]
SALL_PARONLY	0.404	0.114	1e-04	0.389	0.283	[0.20, 0.37]
CENT_PARONLY	0.377	0.112	1e-04	0.398	0.221	[0.15, 0.29]
SALL_PC	0.285	0.077	1e-04	0.283	0.442	[0.35, 0.53]
CENT_PC	0.298	0.067	1e-04	0.310	0.283	[0.20, 0.36]

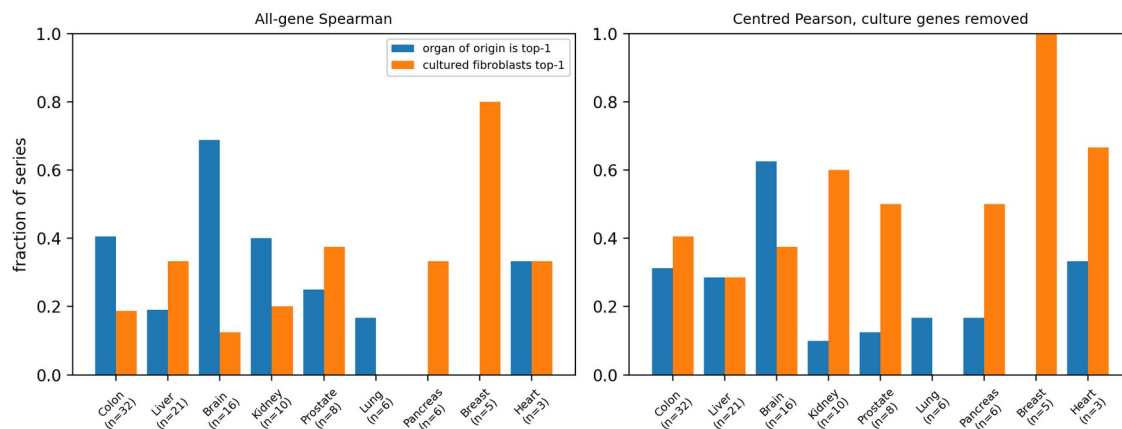


Figure 2. Per-organ share of clean-subset series whose organ of origin (blue) or GTEx cultured fibroblasts (orange) is the top-1 match. Organs with $n \geq 3$.

These series-level ranks were computed before the sample-level audit in Section 4.9. At this stage, the clean text-filtered subset appeared to match cultured fibroblasts often, and the working prediction was that a parenchymal-gene restriction would leave a fibroblast best-match share above 15% in that same text-filtered subset. That prediction is an internal same-series test, not a biological validation. We later checked sample metadata and found many series labeled as organoid-related do not contain organoid samples; Section 4.9 retracts the broad culture-fibroblast interpretation. The tool vorganoid fidelity produces a tissue-rank calculation, not a diagnosis or validated organoid-quality score.

4.7 Purity-corrected fidelity (Human Protein Atlas)

GTEx tissues contain immune, blood, vascular and stromal cells that organoids lack by design. Using the Human Protein Atlas single-cell type atlas, we removed 7,595 genes enhanced in any such cell type (NONPAR), or kept only the 9,699 genes enhanced solely in parenchymal types (PARONLY). Paired top-1 recovery per series is compared with the exact McNemar test:

$$p_{McN} = 2 * \sum_{k=0}^{\min(b,c)} C(b+c, k) 0.5^{(b+c)}, \quad b = \text{base-only hits}, \quad c = \text{variant-only hits} \quad (15)$$

Table 18. Organ-of-origin recovery on 151 series, base method vs gene-restricted variant (results/geo_mcnemar.json).

comparison	top-1 base	top-1 variant	b	c	exact p
SALL vs SALL_NO CULT	0.331	0.331	1	1	1
SALL vs	0.331	0.417	3	16	0.0044

SALL_NO NPAR					
SALL vs SALL_PA RONLY	0.331	0.404	10	21	0.071
SALL vs SALL_PC	0.331	0.285	16	9	0.23
CENT vs CENT_NO CULT	0.252	0.258	0	1	1
CENT vs CENT_NO NPAR	0.252	0.331	4	16	0.012
CENT vs CENT_PA RONLY	0.252	0.377	4	23	0.00031
CENT vs CENT_PC	0.252	0.298	0	7	0.016

Purity restriction raises recovery (all-gene Spearman 33% -> 42%, $p = 0.0044$; centred Pearson 25% -> 38%, $p = 0.00031$). These same-series tissue-rank shifts suggest cell-composition sensitivity, but mixed sample labels prevent a general organoid-specific correction claim. Removing the selected culture genes did not materially change this rank result. The internal text-filtered-subset prediction (fibroblast best-match share above 15% on parenchymal-only genes) was numerically met before the later sample-label audit: 28% [0.20, 0.37] and 22% [0.15, 0.29]. The lower bound of the second interval touches the threshold; more importantly, a threshold met on mixed samples cannot validate a general organoid phenomenon. The strict organoid-only reanalysis in Section 4.9 takes precedence over this earlier text-filtered aggregate. `vorganoid fidelity --purity` applies the gene restriction to a supplied count matrix without asserting clinical or biological calibration.

4.8 What drives the text-filtered fibroblast rank pattern?

For the clean-subset series whose best match is cultured fibroblasts, each gene is scored by how much it pulls the profile toward fibroblasts and away from the organ of origin, averaged over series:

$$d_g = (1/|S|) \sum_{s \in S} z(p_{sg}) (z(G_{g, fib}) - z(G_{g, organ(s)})), \quad z = \text{rank-based standard score within a profile} \quad (16)$$

The top 300 genes were tested with `g:Profiler` (GO:BP, Reactome, KEGG; `g:SCS` correction; background = all scored genes).

Table 19. Enrichment of attractor-driving genes (results/attractor_gprofiler.csv; terms with more than 2,000 genes omitted).

source	term	adjusted p	genes
GO:BP	mitotic cell cycle process	7.8e-56	66
GO:BP	mitotic cell cycle	1.4e-54	69
GO:BP	cell cycle process	3.6e-53	77
GO:BP	cell cycle	1.3e-51	83
GO:BP	adaptive immune response	9.5e-15	31
GO:BP	regulation of immune system process	1.2e-14	43
REAC	Adaptive Immune System	9.0e-13	29

Two components appear: mitotic cell-cycle genes (TOP2A, MKI67, CDK1) that cultures express and tissues do not, and immune and complement genes that tissues contain and these mixed series profiles lack. Removing either family alone leaves the earlier text-filtered rank pattern in place (Sections 4.6-4.7), but the later strictly organoid-only audit in Section 4.9 overrides a general attractor claim. Restricting to 19,254 HGNC protein-coding genes also retains that text-filtered rank pattern (fibroblast best match 44% [0.35, 0.53] and 28% [0.20, 0.36]) in the text-filtered subset; non-coding genes alone do not explain that aggregate, but sample composition still can. A future proliferation/purity hypothesis would need organoid and matched primary tissue from the same donors, profiled together. It cannot be inferred as the mechanism from these mixed series.

4.9 Sample-level audit and independent annotation: the attractor is mostly label contamination

Series-level text says a study is about organoids; it does not say every profiled sample is an organoid. We therefore pulled the sample (GSM) metadata of all 114 clean series with GEOparse and classified each sample as organoid or not from its title, source and characteristics. Only 45 series have every sample annotated as organoid; 59 have no organoid-annotated sample at all (for example GSE343459 profiles fibroblasts isolated from skin organoids, and GSE287925 profiles LNCaP cells in 2D). We compared the fibroblast-best fraction between the strict and the no-organoid series with Fisher's exact test:

$$p = \sum_{x: P(x) \leq P(a)} \frac{C(K, x) C(N - K, n - x)}{C(N, n)} \quad (17)$$

Table 20. Attractor on strict organoid series vs series without organoid samples (results/geo_fidelity_strict.json).

method	strict n	strict top-1	strict fibroblast-best [95%	no-organoid fibroblast-	Fisher p

			CI]	best	
CENT	45	0.33	0.24 [0.13, 0.38]	0.47	0.025
CENT_NO CULT	45	0.33	0.29 [0.18, 0.42]	0.55	0.0095
CENT_NO NPAR	45	0.49	0.20 [0.09, 0.33]	0.52	0.0011
CENT_PA RONLY	45	0.56	0.11 [0.02, 0.20]	0.29	0.03
CENT_PC	45	0.42	0.22 [0.11, 0.36]	0.33	0.27
S3000	45	0.31	0.13 [0.04, 0.24]	0.31	0.038
SALL	45	0.42	0.09 [0.02, 0.18]	0.38	0.0011
SALL_NO CULT	45	0.42	0.09 [0.02, 0.18]	0.36	0.0012
SALL_NO NPAR	45	0.56	0.09 [0.02, 0.18]	0.28	0.023
SALL_PA RONLY	45	0.53	0.11 [0.02, 0.20]	0.41	0.00081
SALL_PC	45	0.44	0.22 [0.11, 0.36]	0.62	6.1e-05

On the strict subset the attractor shrinks to 0.09-0.29 of series (vs 0.28-0.62); the difference is significant in 10 of 11 method variants. Much of the attractor therefore came from non-organoid samples inside organoid-titled series, not from organoids. We retract it as a general property of organoids; it persists only in a minority of strict series.

Independent annotation of the top 200 driver genes with Enrichr and the Reactome AnalysisService reproduces the g:Profiler picture: Hallmark G2-M Checkpoint ($q = 2.5e-33$), CellMarker 'Cycling Basal Cell Trachea Mouse' ($q = 4e-60$), and Reactome 'Classical antibody-mediated complement activation' ($FDR = 1.4e-14$) together with mitotic cell cycle. The driver genes describe what makes a profile look like cultured fibroblasts (proliferation, missing immune and complement programs); they are not an organoid discovery.

4.10 Which organoids are faithful? Organ, deficits and protocol on the strict subset

Table 21. Organ recovery on the 45 strictly organoid series (SALL_PARONLY; results/strict_organ_recovery.csv).

labelled organ	strict series	top-1 recovery	median rank of own tissue
Colon -	19	0.84	1

Transverse			
Brain - Cortex	9	0.78	1
Liver	6	0.17	22
Kidney - Cortex	4	0.00	14.5
Lung	4	0.00	6.5
Esophagus - Mucosa	1	0.00	6
Stomach	1	0.00	3
Uterus	1	0.00	9

Colon and cortical organoids recover their own tissue (23 of 28 top-1); liver, kidney and lung organoids almost never do (1 of 14; Fisher $p = 3.9e-06$). The grouping was chosen after seeing the table, so this p is descriptive. To see what the unfaithful organoids lack we scored, for each series, the deficit of each gene relative to its labelled tissue and averaged within organ:

$$\text{delta_g} = (1/|S_o|) \sum_{\{s \text{ in } S_o\}} [z(G_{g,o}) - z(p_{sg})] \quad (18)$$

The 100 largest deficits were annotated with STRING functional enrichment and Enrichr PanglaoDB. Liver organoids lack the hepatocyte program (Hepatocytes ($q = 1.7e-31$); STRING Organic acid metabolic process ($q = 4.6e-12$)); kidney organoids lack proximal tubule (Proximal Tubule Cells ($q = 5.7e-22$)); lung organoids lack mainly stroma (Pancreatic Stellate Cells ($q = 1.2e-15$)). Faithful colon and cortical organoids lack mostly stromal, vascular and immune cells (Peritubular Myoid Cells ($q = 1.3e-16$); Satellite Glial Cells ($q = 4.8e-14$)).

A direct mechanism test is inconclusive: the share of HPA parenchymal-only genes among the deficits is only modestly higher in unfaithful organs (OR = 1.45, $p = 0.061$), because lung deficits are stromal (share 0.15) while liver (0.68) and kidney (0.55) deficits are parenchymal.

Protocol. Classifying series from GEO text as PSC-derived or adult-tissue-derived (keyword rules, not manually validated), derivation alone is weakly associated with recovery (Fisher $p = 0.10$). The organ effect holds within each stratum (PSC 7/11 vs 0/5, $p = 0.034$; adult 16/16 vs 0/5, $p = 4.9e-05$) and in a Cochran-Mantel-Haenszel test:

$$\text{OR_MH} = \sum_k (a_{kd} / n_k) / \sum_k (b_{kc} / n_k) \quad (19)$$

(pooled OR = 46 with a 0.5 continuity correction, $p = 5.8e-05$). We name the candidate the metabolic-parenchyma gap: liver and kidney organoids miss their tissue because they lack mature metabolic epithelium, while intestinal and cortical organoids match despite missing stroma. It is falsified if, in new strictly organoid liver or kidney series, hepatocyte and proximal-tubule genes do not account for most of the rank gap. Brain organoids are all PSC-derived, so organ and protocol cannot be separated there; per-organ n is small.

Secretome check (UniProt). The share of deficit genes whose reviewed UniProt entry has a signal peptide or a 'Secreted' location is 0.30-0.57 per organ against 0.17 for 2,000 random expressed genes (all $q \leq 0.001$). Every organoid type lacks secreted proteins, faithful or not, so this does not explain the organ split (negative).

4.11 Pre-registered out-of-sample test on ArrayExpress

Before downloading any new data we committed a pre-registration (results/preregistration_organ_split.md): on organoid bulk profiles outside the GEO scan, liver, kidney and lung organoids should rank their own tissue worse than intestinal and brain organoids, tested by a one-sided Mann-Whitney U at 0.05:

$$U = \sum_{\{i \text{ in } \textit{unfaithful}\}} \sum_{\{j \text{ in } \textit{faithful}\}} [1(r_i > r_j) + 0.5 * 1(r_i = r_j)] \quad (20)$$

A BioStudies/ArrayExpress search gave 18 curated human non-cancer organoid studies with processed files; 11 could be scored (the others had mouse gene IDs, probe IDs only, malformed tables, or single-cell data only).

Table 22. Out-of-sample organoid profiles (results/ae_replication.csv).

accession	organ	rank of own tissue	best GTEx match
E-MTAB-9044	liver	39	Stomach
E-MTAB-12822	kidney	2	Cervix - Endocervix
E-MTAB-9807	liver	37	Stomach
E-MTAB-11221	intestine	1	Colon - Transverse
E-MTAB-9134	intestine	1	Colon - Transverse
E-MTAB-11297	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13901	kidney	1	Kidney - Medulla
E-MTAB-6017	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-13578	intestine	1	Small Intestine - Terminal Ileum
E-MTAB-17066	brain	1	Brain - Hypothalamus
E-MTAB-10037	brain	26	Cells - Cultured fibroblasts

Median rank 19.5 vs 1; top-1 0.25 vs 0.86; one-sided $p = 0.031$. The pre-registered test passes, but the support is weak: there is no lung dataset, both liver datasets are cholangiocyte organoids on microarrays (so cell type and platform are confounded), and both kidney datasets match kidney well (ranks 1 and 2), which contradicts the kidney part of the candidate. The metabolic-parenchyma gap remains a candidate; its kidney component did not replicate.

4.12 Orthogonal checks of the metabolic-parenchyma candidate

GSEApY preranked GSEA used MSigDB Hallmark v2023.2 on all ranked strict-subset deficit genes, with 500 permutations. Positive normalized enrichment score (NES) means higher expression in the reference tissue than the organoids. The mean over six metabolic Hallmarks is 1.96 in liver, 1.95 in kidney, 1.65 in brain, and 0.74 in colon (descriptive Mann-Whitney $p=0.004$; pathways overlap and are not independent). Brain nevertheless has oxidative phosphorylation NES 2.63, despite matching its reference tissue. A metabolic deficit alone cannot explain fidelity (results/strict_gsea_summary.json).

Pre-registered Open Targets disease relevance. We compared each organ's top 100 deficit genes against the top 500 targets of its corresponding organ-disease term. Diagonal dominance in the five-by-five overlap matrix is:

$$D = (1/5) \sum_o M_{oo} - (1/20) \sum_{o \neq d} M_{od} \quad (21)$$

$D=2.6$, permutation $p=0.0005$: the primary test passes (results/deficit_opentargets.json). But liver-specific disease enrichment fails (2/100 own-disease hits, $p=0.62$). The pass is driven more by lung and colon than liver; the few hits per cell limit its use as a discovery claim.

Pre-registered decoupler univariate linear modeling used OmniPath CollecTRI transcription factor regulons and PROGENy footprints on 4,261 shared genes. Only 2/12 tested lineage master factors ranked in the top decile of deficient factors (binomial $p=0.34$); the positive control FAILS. Exploratory PROGENy scores show PI3K higher and JAK-STAT and p53 lower in organoids across all five organs, a shared culture signature rather than an organ-specific mechanism (results/deficit_decoupler.json).

Reference robustness. We replaced GTEx with Expression Atlas Illumina Body Map E-MTAB-513. The pre-registered test passed: deficit-vector Spearman rho 0.57-0.72 across organs, top-100 overlap 52-75. The Open Targets diagonal-dominance rerun also passes ($D=2.1$, permutation $p=0.0026$). Both deficit vectors share the organoid measurements, so this checks reference sensitivity, not independent biological replication (results/deficit_bodymap.json).

ClinPGx pharmacogenes: the pre-registered VIP-or-CPIC gene set is unusable because the source release marks all 25,041 genes as VIP. We retain that failure. A documented, post-registration CPIC-guideline-only deviation finds liver deficits in 21 tested genes (median 0.32 vs 0.03 in other genes, $p=0.0004$), but liver specificity fails ($p=0.23$, only 6 genes shared across all organ vectors). This is a drug-testing caution, not evidence of a liver-specific mechanism (results/deficit_clinpgx.json).

Pre-registered population-constraint and coding-composition checks were negative. In gnomAD v2.1.1, liver deficit genes have median LOEUF 1.256 vs brain 1.107 (one-sided $p=0.263$; lower LOEUF is more loss-of-function constrained). Ensembl REST mapped all fixed top-100 genes per organ: liver 99/100 and brain 100/100 are protein coding

(one-sided Fisher $p=1.0$). Neither check supplies a mechanism; the gene universe is already coding-enriched (results/deficit_gnomad.json; results/deficit_ensembl_biotype.json).

Synthesis. Organoid deficits are reproducible to a reference swap and linked to organ-disease targets in aggregate, but the metabolic-parenchyma gap lacks liver specificity and the kidney finding fails external replication. It remains a falsifiable candidate, not a new discovery or a validated diagnostic tool.

4.13 Independent protein and pathway annotation, single-cell challenge, and prior art

Independent pathway annotation (WikiPathways 2026-09-10, NCBI Gene mapping). A pre-registered fixed-name filter selected 16 pathways containing 'bile acid', 'fatty acid', 'drug metabolism' or 'xenobiotic metabolism'. Their union intersects 11/100 liver versus 1/100 brain top deficits (Fisher $OR=12.24$, $p=0.0025$; test passes). This supports the liver top-deficit pathway label in a second resource, not causation or organoid specificity; the comparison follows earlier exploratory results (results/deficit_wikipathways.json).

Table 23. Further pre-registered annotation checks on fixed 100-gene deficit lists, not independent experiments. Each p is descriptive in an exploratory sequence; results/deficit_goa.json, deficit_kegg.json, deficit_rhea.json.

curated source	selected definition	liver top-100	brain top-100	one-sided Fisher p
EBI GOA	3 direct metabolic BP terms	11	2	0.0092
KEGG REST	9 fixed-title metabolic pathways	7	0	0.0070
Rhea	curated Swiss-Prot reaction mapping	42	10	1.5e-07

Direct GOA BP terms cover 11/100 liver versus 2/100 brain top deficits, while any BP annotation covers 94 and 90 respectively. KEGG REST selected 9 pathways with 218 mapped genes. Rhea reactions mark 42 liver versus 10 brain genes; kidney has 41, so the reaction signal is not specific to liver. GOA, KEGG, Rhea, UniProt and WikiPathways annotations may share literature and curation, and prior g:Profiler work included KEGG. These positives cannot be counted as independent confirmations. Neither organoid protein nor enzyme activity was measured, and the original kidney fidelity candidate failed external replication. No mechanism, discovery, or diagnostic use follows from these annotation contrasts.

Additional pre-registered annotation/coverage checks: IntAct PSQUIC queried the fixed top-20 deficit genes per group and measured 20 liver and 19 brain genes. Median indexed interaction RECORD counts were 6.5 versus 45.0 (one-sided rank-test $p=0.0369$). Study density, nonhuman interactors, and blood/immune genes confound this contrast; no organoid interactome was measured. JASPAR CORE vertebrate matrices cover 2/3 pre-fixed brain lineage TFs absent from CollecTRI (NEUROD2 and TBR1, not NEUROD6) and kidney PAX2; this is motif availability only and cannot repair the failed decoupler activity control. Neither is an independent biological replication (results/deficit_intact.json; results/deficit_jaspar.json).

Adult-tissue protein check (HPA v25.1 mass spectrometry). Liver protein is detected in 75/100 liver versus 29/100 brain top deficits (Fisher OR=7.34, $p=4.5e-11$; pre-registered H1 passes). Only 26 liver and 22 brain list proteins have measured intensity in both adult liver and cortex, below the registered 50-per-list minimum, so the quantitative H2 is uninterpretable. There is no organoid proteome in this check (results/deficit_hpa_ms.json).

Single-cell challenge on an independent public human kidney-organoid accession, GSE108291. Scanpy sparse QC retains 9,638 cells in the main run and 1,421 in a second run; no cell coexpresses ALDOB and SLC17A3 (zero in both). The registered 1% two-marker H1 fails; the <20% H2 passes. This does not show that proximal-tubule cells are absent, because sparse single-cell dropout and developmental state remain untested. QC thresholds were set after registration but before looking at marker coexpression; both runs count as one accession (results/sc_kidney_markers.json).

Ontology audit (EBI OLS4) distinguishes kidney cortex UBERON:0001225 from kidney UBERON:0002113, and cholangiocyte CL:1000488 from hepatocyte CL:0000182. This is a label check only, not proof of anatomical hierarchy or expression similarity; the exact 'proximal tubule cell' query was unresolved (results/ols_anatomy_audit.json).

HPO direct gene-phenotype annotation could not test liver-vs-brain metabolic phenotype enrichment: only 35/100 liver and 28/100 brain deficits have any HPO annotation, below the pre-registered 50/list coverage floor. No clinical inference from the missing annotations (results/deficit_hpo.json).

InterPro domain audit is invalid as a top-deficit comparison. Its registered six liver pharmacogenes were incorrectly called top-100 deficits: only CYP2D6 actually is one. The protein-domain annotations remain correct, but the primary contrast cannot be scored (results/deficit_interpro.json).

A fixed three-query Europe PMC title/abstract search returned 6 distinct records. None explicitly reported human intestinal CF organoid starting-size moderation of modulator-induced FIS in its abstract, but absence in a narrow search does not establish novelty.

Related work links nasal organoid lumen to baseline CFTR function and pig pancreatic organoid size to absent lumen (results/epmc_priorart_review.md).

Wider full-text prior art narrows the size claim further: Calucho et al. 2021 tested individual WT nasospheroid starting size against CFTR-dependent FSK shrinking and found no significant association among 138 responders (Spearman $r=-0.1216$, $p=0.1537$ for 60-minute fractional response; supplementary slope $r=0.04124$, $p=0.6298$). This airway system has opposite polarity/readout and does not directly duplicate donor-matched intestinal drug-by-size interaction, but starting-size analysis of CFTR spheroids is not generally novel. Kim et al. 2020 found no initial-size/growth relation in two colorectal tumor organoid lines and warned that small-object detection can add variation. Full-text audit and source URLs: results/epmc_priorart_review.md and results/external_fis_source_audit.md. No discovery follows from a narrower question alone.

Berical et al. 2022 tracked individual human iPSC-derived airway spheroids after forskolin or vehicle and tested CFTR-modulator FIS across patient lines. This is strong prior art for individual-object CFTR swelling assays, not a release of matched individual-object baseline/end size by patient and treatment: its public Source Data workbook provides figure-level experiment values and separate baseline sizes without object IDs linking the panels. Airway tissue and a 20-24-hour response differ from intestinal FIS. Our narrower intestinal drug-by-size result remains unvalidated, not novel merely because this released workbook lacks paired tracks (results/berical_source_suitability.md).

A 2021 Cells study also tracked individual iPSC-derived human intestinal organoids every 15 minutes through a four-hour CFTR swelling assay, measuring size against baseline; it tested VX-809/VX-770 rescue in a single named F508del CF line. Its public supplemental archive contains figures, not raw object tracks, and its underlying data are available on request. Thus even individual tracking of intestinal CFTR FIS predates this work. It cannot establish or refute a donor-general drug-by-starting-size effect because it reports only one CF line and no public joinable object-level treatment table (results/ipsc_intestinal_source_suitability.md).

Synthesis. The liver top-deficit list has independent pathway and adult-tissue protein support, while no new organoid proteome, matched tissue, perturbation, or clinical outcomes validate its effect on fidelity. The kidney external profiles recover their tissue despite the strict single-cell marker test failing. The candidate remains unproven; do not offer the tool as diagnosis.

External replication source audit (results/external_fis_source_audit.md): a 2025 respiratory FIS study with single-organoid measures excluded baseline areas below 1500 px and used forskolin alone rather than modulator vs DMSO; OrganoID's open individual time series used pancreatic cancer organoids and gemcitabine; two nasal CFTR-

modulator papers reported either well-summed areas or summary plots without a verified public matched-object table. Drevinek's 20-patient area series was suitable for well-level drug-ranking analysis, not individual-size replication. This search is not exhaustive.

5. Findings that did not hold

The original sign-only donor-block test (9/12, $p=0.073$) and zero-forskolin specificity test (3/11, $p=0.967$) failed; the later size mixed-model result is same-accession evidence, not donor-independent validation or a CFTR-specific mechanism. The initial power-law interpretation and size-adjusted donor discrimination claim were not retained. The first segmentation checkpoint fell below the published mean, and the v2 evaluation-file AP50 crossed its rounded comparison only after an image split later shown to share organoid fields; it is not an independent benchmark improvement. Full negative-arm results and protocols remain in the repository results and notes, and in Sections 4.3-4.5.

Fidelity claims narrowed after their controls: the broad culture-fibroblast attractor was retracted for strictly organoid-only samples, kidney external profiles contradicted the original organ split, and several mechanism/clinical-target enrichment arms did not meet their registered bars. We retain those numerical tests in the repository rather than expanding each failed arm into the paper's main narrative. Neither the size result nor tissue-rank analysis is a clinical diagnostic.

6. Discussion

The same-accession eligibility sensitivity weakens from 9/12 ($p=0.073$) at three measured objects per cell to 8/12 ($p=0.194$) at five. At ten, the apparently low p comes from only six surviving donors and is uninterpretable under the frozen minimum-eight-donor rule. All three originally nonpositive donors are excluded at minimum ten, so the six-of-six result selects a different donor subset; excluded donors are not imputed as negatives. Selective eligibility is another reason not to claim a donor-general effect (results/finding_block_eligibility_sensitivity.md; results/finding_eligibility_composition.md).

A frozen paired no-forskolin specificity test also fails: 3/11 donor-median contrasts are more positive at 0.128 μM than zero forskolin (one-sided $p=0.9673$). This weakens the proposed CFTR-specific interpretation without proving its absence or explaining the zero-dose response (results/finding_zero_fsk_control.md).

In this accession, drug-minus-DMSO swelling tends to rise from small to larger organoids and then plateau. The original plate-and-dose-matched donor sign test fails at alpha 0.05, while the later locked magnitude-aware donor test and mixed model detect a same-accession positive interaction. This conflict of estimator sensitivity does not establish an independently replicated or CFTR-specific donor-general effect. Uniform surface-flux geometry predicts the reverse direction, but that simple model is not a full

measurement-bias test. Lumen maturity and cell composition are hypotheses, not measured explanations. Falsifiable prediction: in an independent, matched, time-lapse single-organoid intestinal FIS cohort, with size thresholds locked before analysis, a donor-level modulator-minus-DMSO large-minus-small effect will be positive in most donors, with exact donor-magnitude and mixed-model tests agreeing in a new cohort. This needs a second suitable dataset and mechanistic measurements; the public leads audited here do not supply those controls.

Practical note. In this dataset, starting size is associated with some exploratory single-organoid readouts, but the registered matched-block and specificity tests fail, and size correction did not improve per-donor discrimination. Recording the starting-size distribution across passages is a possible research quality-control variable, not a tested intervention or clinical adjustment.

Independent novelty and mechanism remain unproved. We did not find the exact intestinal drug-by-starting-size interaction in the bounded literature search, but Calucho et al. 2021 tested starting size versus individual CFTR nasospheroid FSK response and found no association. Berical et al. 2022 tracked individual iPSC-derived airway spheroid FIS and tested modulators across multiple patient lines; the released source workbook does not link patient-treatment outcomes to per-object baseline areas. A 2021 Cells study had already tracked individual iPSC-derived intestinal CFTR-swelling objects in one CF line; its public supplement lacks paired raw object tracks. A nasal 2D-derived-organoid study reported swelling variation in large structures (bioRxiv 2021.07.20.453105). The narrower intestinal question is unconfirmed and this was not a systematic review.

7. Tools used

Table 24. Tools genuinely used (results/tools_ledger.csv): 47 entries, 40 counting toward the gate after excluding infrastructure. Target of 40 reached; this counts resources genuinely used, not independent biological confirmations.

tool	kind	where used	gate
NumPy	library	src/vorganoid; src/geo_fidelity.py; paper/build_paper.py	counts
SciPy	library	src/geo_fidelity.py (spearmanr); paper/build_paper.py	counts
pandas	library	src/geo_fidelity.py; paper/build_paper.py	counts
scikit-image	library	src/vorganoid	counts

		(instance post-processing; watershed/labels)	
PyTorch	library	train_seg.py; tune_seg.py (U-Net CNN)	counts
Pillow	library	train_seg.py image loading	infra (excluded)
python-docx	library	paper/paperkit.py	infra (excluded)
pytest	test framework	tests/ (hermetic suite)	infra (excluded)
LibreOffice (soffice headless)	application	paper docx -> pdf render	infra (excluded)
poppler pdfinfo	application	page count of rendered paper	infra (excluded)
Git + GitHub	service	repo hosting and pushes	infra (excluded)
NCBI E-utilities (esearch/esummary)	web API	src/geo_candidates.py	counts
NCBI GEO FTP/HTTPS supplementary files	data service	src/geo_fidelity.py	counts
GTEX v8 gene median TPM (GTEX portal bucket)	reference data service	data/ref/gtex_v8_median_tpm.gct.gz; src/geo_fidelity.py	counts
MSigDB Hallmark gene sets v2023.2 (Broad GSEA-MSigDB)	gene-set resource	data/ref/h.all.v2023.2.Hs.symbol.gmt; src/geo_fidelity_methods.py (culture-confound test)	counts
matplotlib	library	src/fig_geo_fidelity.py; paper figures (fig_size_attenuation.png, fig_geo_fidelity.png)	counts
Human Protein Atlas single-cell type API	database/web API	data/ref/hpa_single_cell_type_v_api.tsv.gz; src/geo_fidelity_me	counts

		thods.py (purity gene sets)	
statsmodels	library	src/ geo_mcnemar.py (exact McNemar tests)	counts
g:Profiler g:GOST API (GO, Reactome, KEGG)	web API/database	src/ geo_attractor_genes.py (enrichment of attractor-driving genes)	counts
HGNC complete set (locus groups)	database	data/ref/ hgnc_locus_groups.csv.gz; src/geo_fidelity_methods.py (protein-coding robustness)	counts
GEOparse	library	src/ geo_sample_meta.py	counts
Enrichr (Hallmark/CellMarker/PanglaoDB libraries)	service	src/ attractor_enrichr_reactome.py	counts
Reactome AnalysisService	service	src/ attractor_enrichr_reactome.py	counts
STRING functional enrichment API (string-db.org)	service	src/ strict_organ_deficits.py	counts
UniProt REST (UniProtKB reviewed human)	database	src/ strict_deficit_uniprot.py	counts
EBI BioStudies / ArrayExpress API + FTP	database	src/ae_scan.py; src/ae_replication.py	counts
GSEAPy prerank (v1.3.1)	library	src/strict_gsea.py	counts
Open Targets Platform GraphQL API (v26.9)	database/service	src/ deficit_opentargets.py	counts
decoupler 2.1.4 (ulm)	library	src/ deficit_decoupler.py	counts

		y	
OmniPath (CollectRI TF regulons + PROGENy pathway footprints)	database/service	src/ deficit_decoupler.p y	counts
EBI Expression Atlas (baseline experiments FTP)	database/service	src/ deficit_bodymap.py	counts
ClinPGx / PharmGKB gene download (release 2026-09-05)	database/service	src/ deficit_clinpgx.py	counts
gnomAD v2.1.1 gene-level pLoF constraint	reference database	src/ deficit_gnomad.py; results/deficit_gno mad.json	counts
Ensembl REST lookup/id	web API	src/ deficit_ensembl_bi otype.py; results/deficit_ense mbl_biotype.json	counts
EBI Ontology Lookup Service 4 (OLS4)	ontology web API	src/ ols_anatomy_audit. py; results/ols_anatomy .json	counts
Scanpy 1.11.5	library	src/ sc_kidney_markers. py; results/sc_kidney_ markers.json	counts
Human Phenotype Ontology gene-to- phenotype release	reference ontology/database	src/deficit_hpo.py; results/deficit_hpo.j son	counts
Europe PMC REST search	literature API	src/ epmc_priorart.py; results/epmc_priora rt.json; results/epmc_priora rt_review.md	counts
InterPro protein- domain API	database/service	src/ deficit_interpro.py; results/deficit_inter pro.json	counts

WikiPathways 2026-09-10 human GMT	pathway database	src/ deficit_wikipathwa ys.py; results/deficit_wiki pathways.json	counts
NCBI Gene human gene_info	reference database	src/ deficit_wikipathwa ys.py; results/deficit_wiki pathways.json	counts
Human Protein Atlas mass- spectrometry tissue download	dataset	src/ deficit_hpa_ms.py; results/deficit_hpa_ ms.json	infra (excluded)
EBI GOA human GAF	gene ontology annotation database	src/deficit_goa.py; results/deficit_goa.j son	counts
KEGG REST human pathway- gene API	database/service	src/deficit_kegg.py; results/deficit_kegg .json	counts
Rhea curated biochemical reaction-to- UniProt mapping	reference database	src/deficit_rhea.py; results/deficit_rhea. json	counts
IntAct PSIQUIC interaction-count service	protein interaction database	src/ deficit_intact.py; results/deficit_intac t.json	counts
JASPAR CORE vertebrate motif API	transcription factor motif database	src/ deficit_jaspar.py; results/deficit_jaspa r.json	counts

References

Lefferts JW, et al. OrgaSegment: deep-learning based organoid segmentation to quantify CFTR dependent fluid secretion. Commun Biol 2024. <https://www.nature.com/articles/s42003-024-05966-4>

Boj SF, Vonk AM, et al. Forskolin-induced swelling in intestinal organoids: an in vitro assay for assessing drug response in cystic fibrosis patients. J Vis Exp 2017. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5408767/>

GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science 2020. <https://gtexportal.org>

Karlsson M, et al. A single-cell type transcriptomics map of human tissues. Sci Adv 2021. <https://www.proteinatlas.org>

Szklarczyk D, et al. The STRING database in 2023. Nucleic Acids Res 2023.

The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Res 2023.

Moreno P, et al. Expression Atlas and ArrayExpress/BioStudies at EMBL-EBI. Nucleic Acids Res 2022.

Open Targets Platform. <https://platform.opentargets.org/>

Badia-i-Mompel P, et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. Bioinform Adv 2022. <https://bioconductor.org/packages/decoupleR/>

Turei D, et al. OmniPath: guidelines and gateway for literature-curated signaling pathway resources. Nat Methods 2016. <https://omnipathdb.org/>

ClinPGx / PharmGKB. <https://www.clinpgx.org/>

Karczewski KJ, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. Nature 2020. <https://gnomad.broadinstitute.org/help/constraint>

Ensembl REST API. https://rest.ensembl.org/documentation/info/lookup_post

WikiPathways monthly releases. <https://data.wikipathways.org/20260910/gmt/>

EBI GOA human GAF. https://ftp.ebi.ac.uk/pub/databases/GO/goa/HUMAN/goa_human.gaf.gz

IntAct PSQUIC. <https://www.ebi.ac.uk/Tools/webservices/psicquic/intact/webservices/current/search/query/>

JASPAR CORE vertebrate motif API. <https://jaspar.elixir.no/api/v1/matrix/>

KEGG REST API. <https://www.kegg.jp/kegg/rest/keggapi.html>

Rhea reaction-to-Swiss-Prot mapping. https://ftp.expasy.org/databases/rhea/tsv/rhea2uniprot_sprot.tsv

Human Protein Atlas protein mass-spectrometry normal tissue download. <https://www.proteinatlas.org/humanproteome/tissue/data>

Human Phenotype Ontology gene-to-phenotype annotation. https://obophenotype.github.io/human-phenotype-ontology/annotations/genes_to_phenotype/

EBI Ontology Lookup Service 4. <https://www.ebi.ac.uk/ols4/api-docs>

Human Cell Atlas GSE108291 kidney organoid single-cell project.
<https://explore.data.humancellatlas.org/projects/7b947aa2-43a7-4082-afff-222a3e3a4635>

Demchenko A, et al. A semi-automated algorithm for image analysis of respiratory organoids. *PLoS Comput Biol* 2025. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013589>

Matthews J, et al. OrganoID. *PLoS Comput Biol* 2022.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9645660/>

Anderson JD, et al. CFTR function and clinical response to modulators parallel nasal epithelial organoid swelling. *AJP Lung* 2021.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8321858/>

Calucho M, et al. Validation of nasospheroids to assay CFTR functionality and modulator responses in cystic fibrosis. *Sci Rep* 2021. <https://www.nature.com/articles/s41598-021-94798-x>

Berical A, et al. A multimodal iPSC platform for cystic fibrosis drug testing. *Nat Commun* 2022. <https://www.nature.com/articles/s41467-022-31854-8> . Source-data suitability: results/berical_source_suitability.md

High-Throughput Functional Analysis of CFTR and Other Apically Localized Proteins in iPSC-Derived Human Intestinal Organoids. *Cells* 2021. <https://www.mdpi.com/2073-4409/10/12/3419> . Supplement suitability: results/ipsc_intestinal_source_suitability.md

Kim S, et al. Comparison of Cell and Organoid-Level Analysis of Patient-Derived 3D Organoids to Evaluate Tumor Cell Growth Dynamics and Drug Response. *SLAS Discov* 2020. [https://www.slas-discovery.org/article/S2472-5552\(22\)06605-9/fulltext](https://www.slas-discovery.org/article/S2472-5552(22)06605-9/fulltext)

Botelho H, Hagemeijer MC, et al. FIS_image_analysis demonstration dataset.
https://github.com/hmbotelho/FIS_image_analysis

Drevinek P, et al. Response to elexacaftor/tezacaftor/ivacaftor in intestinal organoids derived from people with cystic fibrosis. *J Cyst Fibros* 2021.
<https://doi.org/10.1016/j.jcf.2021.07.006> . Data: <https://zenodo.org/records/4771466>

Borek-Dohalska L, et al. Effect of vanzacaftor on cystic fibrosis airway epithelial cells compared to elexacaftor. Data: <https://zenodo.org/records/15754800>

InterPro API. <https://interpro-documentation.readthedocs.io/en/latest/api.html>

Europe PMC RESTful Web Service. <https://europepmc.org/RestfulWebService>

Kuleshov MV, et al. Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. *Nucleic Acids Res* 2016.

Milacic M, et al. The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Res* 2024.

Gumienny R. GEOparse: Python library to access Gene Expression Omnibus. <https://github.com/guma44/GEOparse>

Kolberg L, et al. g:Profiler - interoperable web service for functional enrichment analysis and gene identifier mapping (2023 update). Nucleic Acids Res 2023. <https://biit.cs.ut.ee/gprofiler>

Seal RL, et al. Genenames.org: the HGNC resources in 2023. Nucleic Acids Res 2023. <https://www.genenames.org>

Liberzon A, et al. The Molecular Signatures Database Hallmark gene set collection. Cell Syst 2015. <https://www.gsea-msigdb.org>

Barrett T, et al. NCBI GEO: archive for functional genomics data sets. Nucleic Acids Res 2013. <https://www.ncbi.nlm.nih.gov/geo/>

Botelho HM. FIS_analysis. https://github.com/hmbotelho/FIS_analysis

Measuring cystic fibrosis drug responses in organoids derived from 2D differentiated nasal epithelia. bioRxiv 2021. <https://www.biorxiv.org/content/10.1101/2021.07.20.453105v2>

Appendix A. Patient, plate and dose-level treatment contrasts

This appendix makes the two distinct well-level FIS rankings inspectable without downloading a raw table. Every row is one source patient x plate x forskolin-dose block, not a new patient or accession. Effects average technical wells within an arm before the patient median is formed. Normalized area AUC is dimensionless and divided by the observation horizon; values here are differences of treatment-arm AUCs. These results cannot test individual-organoid size and should not be used to choose therapy.

A1 uses Zenodo 4771466 and triple ELX/TEZ/IVA minus double TEZ/IVA. A2 uses Zenodo 15754800 and default VTI minus ETI. The source releases come from related teams, and anonymized patient numbers may overlap. The two experiments, drugs and eligibility rules are not pooled. See results/preregistration_drevinek_fis.md and results/preregistration_vti_eti.md for the frozen questions; the source scripts and JSON contain every eligible well, time-point count, exclusion and sensitivity test.

A1. ELX/TEZ/IVA minus TEZ/IVA

Source <https://zenodo.org/records/4771466>. 20 IDs and 54 plates appear in the public source; 431/432 blocks meet its preregistered primary criteria. The table preserves positive and negative contrasts. The registered patient-level result is 20/20 positive medians, exact one-sided $p=9.536743e-07$. Unnormalized raw-area AUC is shown only to reveal sensitivity to baseline area, never as an independent trial.

A1.1. Source patient 04-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2397; median raw-area contrast +80.78. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 25. A1 source patient 04-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-20	0.008192	+0.0969	+17.2
2020-08-20	0.02048	+0.1733	-25.0
2020-08-20	0.0512	+0.2351	+105.8
2020-08-20	0.128	+0.3862	+84.2
2020-08-20	0.32	+0.3403	+109.7
2020-08-20	0.8	+0.4299	+72.8
2020-08-20	2	+0.4613	+84.1
2020-08-20	5	+0.4579	+135.0
2020-08-28	0.008192	+0.1165	-3.5

2020-08-28	0.02048	+0.1137	+65.2
2020-08-28	0.0512	+0.1571	+50.4
2020-08-28	0.128	+0.2123	+77.5
2020-08-28	0.32	+0.1809	+89.4
2020-08-28	0.8	+0.3861	+126.2
2020-08-28	2	+0.4189	+101.3
2020-08-28	5	+0.4469	+104.3
2020-09-03	0.008192	+0.0601	+31.4
2020-09-03	0.02048	+0.1244	+56.3
2020-09-03	0.0512	+0.1790	+59.4
2020-09-03	0.128	+0.3319	+101.6
2020-09-03	0.32	+0.3905	+28.3
2020-09-03	0.8	+0.3044	+143.4
2020-09-03	2	+0.2444	-7.7
2020-09-03	5	+0.1897	+88.8

A1.2. Source patient 05-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2443; median raw-area contrast +69.15. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 26. A1 source patient 05-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-11-19	0.008192	+0.0627	+58.1
2020-11-19	0.02048	+0.1807	+68.7
2020-11-19	0.0512	+0.2492	+135.9
2020-11-19	0.128	+0.1121	+123.0
2020-11-19	0.32	+0.1348	+58.2
2020-11-19	0.8	+0.2862	+128.1
2020-11-19	2	+0.2863	+195.0
2020-11-19	5	+0.1021	-7.7
2020-11-27	0.008192	+0.1676	+69.6
2020-11-27	0.02048	+0.1739	+59.6
2020-11-27	0.0512	+0.3493	+149.2
2020-11-27	0.128	+0.3110	+75.5
2020-11-27	0.32	+0.1827	+14.2
2020-11-27	0.8	+0.2861	+237.0
2020-11-27	2	+0.2918	+51.5
2020-11-27	5	+0.4184	+241.6
2020-12-04	0.008192	+0.0599	+34.4

2020-12-04	0.02048	+0.3061	+25.6
2020-12-04	0.0512	+0.1850	+113.5
2020-12-04	0.128	+0.3075	+76.1
2020-12-04	0.32	+0.2394	+1.4
2020-12-04	0.8	+0.3160	+54.4
2020-12-04	2	+0.2506	+80.9
2020-12-04	5	+0.2039	+24.9

A1.3. Source patient 10-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2502; median raw-area contrast +60.43. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 27. A1 source patient 10-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-11-19	0.008192	+0.2087	+93.1
2020-11-19	0.02048	+0.1557	+69.1
2020-11-19	0.0512	+0.3545	+45.9
2020-11-19	0.128	+0.4348	+82.3
2020-11-19	0.32	+0.3166	+35.9
2020-11-19	0.8	+0.2305	+40.0
2020-11-19	2	+0.1722	+61.9
2020-11-19	5	+0.2287	+13.6
2020-11-25	0.008192	+0.1976	+20.9
2020-11-25	0.02048	+0.2663	+119.8
2020-11-25	0.0512	+0.3775	+122.3
2020-11-25	0.128	+0.4570	+123.5
2020-11-25	0.32	+0.3966	+50.4
2020-11-25	0.8	+0.4991	+84.5
2020-11-25	2	+0.3223	+1.1
2020-11-25	5	+0.1941	+25.3
2021-01-06	0.008192	+0.2091	+26.5
2021-01-06	0.02048	+0.2342	+95.2
2021-01-06	0.0512	+0.1848	+13.6
2021-01-06	0.128	+0.4305	+109.8
2021-01-06	0.32	+0.2299	+59.0
2021-01-06	0.8	+0.4818	+100.7
2021-01-06	2	+0.3073	-0.4
2021-01-06	5	+0.2228	+108.4

A1.4. Source patient 20-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.1534; median raw-area contrast +108.65. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 28. A1 source patient 20-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1287	+135.3
2020-08-27	0.02048	+0.1368	+226.3
2020-08-27	0.0512	+0.1487	+129.1
2020-08-27	0.128	-0.0330	+123.2
2020-08-27	0.32	+0.1130	+171.5
2020-08-27	0.8	-0.1799	-12.5
2020-08-27	2	-0.0607	+49.3
2020-08-27	5	+0.0265	+133.3
2020-09-08	0.008192	+0.0891	-1.6
2020-09-08	0.02048	+0.1368	-13.8
2020-09-08	0.0512	+0.2866	+103.0
2020-09-08	0.128	+0.1152	+105.9
2020-09-08	0.32	+0.2096	+24.0
2020-09-08	0.8	+0.3125	+104.0
2020-09-08	2	+0.2647	+110.8
2020-09-08	5	+0.1582	-44.8
2020-09-15	0.008192	+0.1473	+97.1
2020-09-15	0.02048	+0.2262	+160.2
2020-09-15	0.0512	+0.3844	+157.1
2020-09-15	0.128	+0.3246	+191.7
2020-09-15	0.32	+0.3829	+133.5
2020-09-15	0.8	+0.3559	+106.5
2020-09-15	2	+0.3542	+14.2
2020-09-15	5	+0.1749	+155.8

A1.5. Source patient 21-CF

23 eligible of 24 blocks; median normalized treatment contrast +0.2527; median raw-area contrast +97.31. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 29. A1 source patient 21-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1814	+62.9
2020-08-13	0.02048	+0.1679	-10.0
2020-08-13	0.0512	+0.3067	+97.3
2020-08-13	0.128	+0.4494	+200.0
2020-08-13	0.32	+0.9694	+126.1
2020-08-13	0.8	+0.5938	+69.2
2020-08-13	2	+0.7810	+220.4
2020-08-13	5	+0.8583	+217.4
2020-09-18	0.008192	+0.1843	-23.7
2020-09-18	0.02048	excluded	<2 wells/arm
2020-09-18	0.0512	+0.2527	+94.5
2020-09-18	0.128	+0.1219	+42.9
2020-09-18	0.32	+0.1303	-51.0
2020-09-18	0.8	+0.2095	+145.0
2020-09-18	2	+0.0519	+130.2
2020-09-18	5	+0.0882	-37.7
2020-09-24	0.008192	+0.2207	+127.9
2020-09-24	0.02048	+0.3608	+125.7
2020-09-24	0.0512	+0.2756	+113.1
2020-09-24	0.128	+0.2576	+229.0
2020-09-24	0.32	+0.3469	+42.5
2020-09-24	0.8	+0.3861	+109.3
2020-09-24	2	+0.1227	+38.6
2020-09-24	5	-0.0843	+59.5

A1.6. Source patient 22-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2165; median raw-area contrast +87.72. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 30. A1 source patient 22-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1680	+61.0
2020-08-27	0.02048	+0.1146	+60.2
2020-08-27	0.0512	+0.2120	+191.8
2020-08-27	0.128	+0.3208	+128.2
2020-08-27	0.32	+0.3724	+219.8
2020-08-27	0.8	+0.3202	+94.8

2020-08-27	2	+0.3405	+190.0
2020-08-27	5	+0.0314	-137.4
2020-09-01	0.008192	+0.1006	+86.2
2020-09-01	0.02048	+0.1493	-4.5
2020-09-01	0.0512	+0.1727	+15.0
2020-09-01	0.128	+0.1303	+109.8
2020-09-01	0.32	+0.1500	+9.9
2020-09-01	0.8	+0.3940	+72.0
2020-09-01	2	+0.2823	+180.1
2020-09-01	5	+0.0797	+91.7
2020-09-24	0.008192	+0.1247	+87.8
2020-09-24	0.02048	+0.2495	+150.2
2020-09-24	0.0512	+0.2776	+49.7
2020-09-24	0.128	+0.2209	+87.7
2020-09-24	0.32	+0.2853	-67.6
2020-09-24	0.8	+0.4043	+241.1
2020-09-24	2	+0.1834	+51.3
2020-09-24	5	+0.2237	+104.2

A1.7. Source patient 41-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3282; median raw-area contrast +71.25. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 31. A1 source patient 41-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1753	+22.4
2020-08-13	0.02048	+0.2519	+114.0
2020-08-13	0.0512	+0.3822	+88.3
2020-08-13	0.128	+0.5059	+119.7
2020-08-13	0.32	+0.5738	+72.4
2020-08-13	0.8	+0.8446	+224.9
2020-08-13	2	+0.4962	+161.0
2020-08-13	5	+0.6016	+126.1
2020-08-20	0.008192	+0.1145	-1.7
2020-08-20	0.02048	+0.1807	+32.8
2020-08-20	0.0512	+0.2720	+32.3
2020-08-20	0.128	+0.4330	+60.2
2020-08-20	0.32	+0.5062	+93.3
2020-08-20	0.8	+0.5894	+128.8

2020-08-20	2	+0.3326	+50.8
2020-08-20	5	+0.3239	+7.4
2020-08-28	0.008192	+0.1596	+70.1
2020-08-28	0.02048	+0.1603	+49.8
2020-08-28	0.0512	+0.4934	+111.1
2020-08-28	0.128	+0.3831	+99.3
2020-08-28	0.32	+0.2534	+51.4
2020-08-28	0.8	+0.1989	-83.3
2020-08-28	2	+0.1937	-107.3
2020-08-28	5	+0.2143	+169.7

A1.8. Source patient 47-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3002; median raw-area contrast +99.63. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 32. A1 source patient 47-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.2497	+99.1
2020-08-27	0.02048	+0.2530	+64.5
2020-08-27	0.0512	+0.3126	+74.7
2020-08-27	0.128	+0.3431	+135.7
2020-08-27	0.32	+0.5456	+165.3
2020-08-27	0.8	+0.2775	+42.5
2020-08-27	2	+0.4075	-12.4
2020-08-27	5	+0.3058	+217.4
2020-11-05	0.008192	+0.2353	+26.0
2020-11-05	0.02048	+0.1838	+133.4
2020-11-05	0.0512	+0.3257	+99.7
2020-11-05	0.128	+0.3508	+60.3
2020-11-05	0.32	+0.2867	+28.5
2020-11-05	0.8	+0.3139	+209.5
2020-11-05	2	+0.1556	+99.6
2020-11-05	5	+0.2898	+79.0
2020-11-24	0.008192	+0.0862	-34.1
2020-11-24	0.02048	+0.2555	+111.2
2020-11-24	0.0512	+0.4055	+182.2
2020-11-24	0.128	+0.2708	+97.8
2020-11-24	0.32	+0.3053	+105.9
2020-11-24	0.8	+0.3092	+123.7

2020-11-24	2	+0.3773	+187.2
2020-11-24	5	+0.2950	+148.1

A1.9. Source patient 57-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2752; median raw-area contrast +75.66. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 33. A1 source patient 57-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-27	0.008192	+0.1154	-51.2
2020-08-27	0.02048	+0.1659	+76.0
2020-08-27	0.0512	+0.2875	+75.3
2020-08-27	0.128	+0.3663	+186.3
2020-08-27	0.32	+0.3707	-19.4
2020-08-27	0.8	+0.0961	+33.6
2020-08-27	2	+0.5050	+375.7
2020-08-27	5	+0.5034	+273.6
2020-09-01	0.008192	+0.1516	+94.5
2020-09-01	0.02048	+0.2158	+137.0
2020-09-01	0.0512	+0.1955	+138.9
2020-09-01	0.128	+0.2320	+1.5
2020-09-01	0.32	+0.3394	+142.5
2020-09-01	0.8	+0.1961	+103.9
2020-09-01	2	+0.3137	+313.3
2020-09-01	5	+0.5433	+130.7
2020-09-08	0.008192	+0.1598	+8.1
2020-09-08	0.02048	+0.1472	+46.1
2020-09-08	0.0512	+0.3105	+72.0
2020-09-08	0.128	+0.2637	+65.4
2020-09-08	0.32	+0.2097	-37.6
2020-09-08	0.8	+0.2868	+61.1
2020-09-08	2	+0.3960	+120.1
2020-09-08	5	+0.3423	+36.4

A1.10. Source patient 58-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.0189; median raw-area contrast +33.19. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 34. A1 source patient 58-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-09-25	0.008192	-0.0068	-6.1
2020-09-25	0.02048	+0.1879	+163.8
2020-09-25	0.0512	+0.1724	+109.7
2020-09-25	0.128	+0.0598	+46.9
2020-09-25	0.32	-0.0104	-102.7
2020-09-25	0.8	-0.1529	+40.6
2020-09-25	2	-0.1628	-10.0
2020-09-25	5	-0.1372	+29.8
2020-10-13	0.008192	+0.1115	+89.5
2020-10-13	0.02048	+0.1405	+89.1
2020-10-13	0.0512	+0.0679	+7.2
2020-10-13	0.128	-0.0182	+49.3
2020-10-13	0.32	+0.3089	+95.3
2020-10-13	0.8	-0.1031	+19.0
2020-10-13	2	-0.0962	+14.2
2020-10-13	5	-0.0557	-1.8
2020-11-20	0.008192	+0.1027	-23.4
2020-11-20	0.02048	+0.0586	+42.0
2020-11-20	0.0512	+0.0210	+8.3
2020-11-20	0.128	+0.0302	+57.1
2020-11-20	0.32	+0.0169	-7.3
2020-11-20	0.8	-0.0293	+12.5
2020-11-20	2	+0.0856	+63.2
2020-11-20	5	-0.0563	+36.6

A1.11. Source patient 59-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.3739; median raw-area contrast +90.31. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 35. A1 source patient 59-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-13	0.008192	+0.1418	+45.9
2020-08-13	0.02048	+0.2441	+122.3
2020-08-13	0.0512	+0.2658	+89.4

2020-08-13	0.128	+0.5715	+81.5
2020-08-13	0.32	+0.5435	+19.1
2020-08-13	0.8	+0.6986	+121.4
2020-08-13	2	+0.5563	+75.5
2020-08-13	5	+0.4616	+87.6
2020-08-21	0.008192	+0.1856	+95.5
2020-08-21	0.02048	+0.2447	+67.3
2020-08-21	0.0512	+0.3450	+165.6
2020-08-21	0.128	+0.4972	+142.5
2020-08-21	0.32	+0.4028	+154.3
2020-08-21	0.8	+0.4609	+266.4
2020-08-21	2	+0.4136	+117.9
2020-08-21	5	+0.4991	+229.6
2020-08-28	0.008192	+0.1663	+44.8
2020-08-28	0.02048	+0.1817	+77.2
2020-08-28	0.0512	+0.1962	+91.2
2020-08-28	0.128	+0.3427	+56.7
2020-08-28	0.32	+0.1733	+86.7
2020-08-28	0.8	+0.4969	+146.9
2020-08-28	2	+0.5447	+204.8
2020-08-28	5	+0.2635	+4.4

A1.12. Source patient 60-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2855; median raw-area contrast +73.90. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 36. A1 source patient 60-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-08-28	0.008192	+0.0823	+25.4
2020-08-28	0.02048	+0.0338	+69.0
2020-08-28	0.0512	+0.2529	+12.3
2020-08-28	0.128	+0.2634	+79.1
2020-08-28	0.32	+0.2767	+98.3
2020-08-28	0.8	+0.3716	+62.2
2020-08-28	2	+0.2134	+90.7
2020-08-28	5	+0.3016	+58.9
2020-09-03	0.008192	+0.0974	+69.7
2020-09-03	0.02048	+0.2117	+118.4
2020-09-03	0.0512	+0.2263	+29.4

2020-09-03	0.128	+0.3349	+83.3
2020-09-03	0.32	+0.2943	+82.1
2020-09-03	0.8	+0.4011	+101.6
2020-09-03	2	+0.3729	+196.5
2020-09-03	5	+0.4666	+158.3
2021-02-17	0.008192	+0.1030	+50.1
2021-02-17	0.02048	+0.1163	+32.8
2021-02-17	0.0512	+0.2100	+55.7
2021-02-17	0.128	+0.3205	+59.3
2021-02-17	0.32	+0.4397	+165.4
2021-02-17	0.8	+0.3940	+107.2
2021-02-17	2	+0.3039	+66.6
2021-02-17	5	+0.3358	+78.1

A1.13. Source patient 61-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.2065; median raw-area contrast +40.30. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 37. A1 source patient 61-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-14	0.008192	+0.1010	+3.4
2020-10-14	0.02048	+0.1303	+61.6
2020-10-14	0.0512	+0.2713	+94.0
2020-10-14	0.128	+0.3055	+1.0
2020-10-14	0.32	+0.1864	+62.9
2020-10-14	0.8	+0.2266	+9.0
2020-10-14	2	+0.2298	+33.6
2020-10-14	5	+0.1652	+47.0

A1.14. Source patient 62-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2456; median raw-area contrast +65.39. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 38. A1 source patient 62-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
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2020-10-13	0.008192	+0.1148	+59.2
2020-10-13	0.02048	+0.1510	+134.3
2020-10-13	0.0512	+0.2029	+99.3
2020-10-13	0.128	+0.2730	+3.1
2020-10-13	0.32	+0.2679	+112.1
2020-10-13	0.8	+0.3107	+190.4
2020-10-13	2	+0.2862	+198.4
2020-10-13	5	+0.2334	+35.7
2020-10-23	0.008192	+0.1637	+9.4
2020-10-23	0.02048	+0.2179	+164.0
2020-10-23	0.0512	+0.2163	+96.9
2020-10-23	0.128	+0.2786	+60.1
2020-10-23	0.32	+0.2270	+2.8
2020-10-23	0.8	+0.2539	+122.8
2020-10-23	2	+0.3275	+161.1
2020-10-23	5	+0.3026	-28.4
2020-11-03	0.008192	+0.1588	+70.7
2020-11-03	0.02048	+0.1511	+57.5
2020-11-03	0.0512	+0.1913	+56.3
2020-11-03	0.128	+0.2564	+169.0
2020-11-03	0.32	+0.2374	+118.7
2020-11-03	0.8	+0.2807	-52.1
2020-11-03	2	+0.2631	+14.4
2020-11-03	5	+0.2728	-7.6

A1.15. Source patient 63-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.1083; median raw-area contrast +35.44. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 39. A1 source patient 63-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-14	0.008192	+0.0993	-54.9
2020-10-14	0.02048	+0.1585	+84.2
2020-10-14	0.0512	+0.1628	+56.2
2020-10-14	0.128	+0.0816	+34.7
2020-10-14	0.32	+0.0410	+29.7
2020-10-14	0.8	+0.1787	+36.2
2020-10-14	2	+0.0252	+4.6
2020-10-14	5	+0.1172	+49.2

A1.16. Source patient 64-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2636; median raw-area contrast +72.83. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 40. A1 source patient 64-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-02	0.008192	+0.1263	+68.9
2020-10-02	0.02048	+0.2773	+156.1
2020-10-02	0.0512	+0.3902	+61.7
2020-10-02	0.128	+0.3963	-61.1
2020-10-02	0.32	+0.3450	+190.4
2020-10-02	0.8	+0.1578	+87.4
2020-10-02	2	+0.8928	+411.1
2020-10-02	5	+0.1912	+28.6
2020-10-13	0.008192	+0.1390	+10.9
2020-10-13	0.02048	+0.1920	+52.1
2020-10-13	0.0512	+0.2484	+132.3
2020-10-13	0.128	+0.4522	+160.1
2020-10-13	0.32	+0.3708	+94.7
2020-10-13	0.8	+0.3116	+132.1
2020-10-13	2	+0.2478	+76.8
2020-10-13	5	+0.1562	+62.1
2020-10-23	0.008192	+0.1398	+36.0
2020-10-23	0.02048	+0.1458	+45.2
2020-10-23	0.0512	+0.2500	+7.5
2020-10-23	0.128	+0.3986	+145.2
2020-10-23	0.32	+0.2824	+18.9
2020-10-23	0.8	+0.4767	+133.5
2020-10-23	2	+0.6337	+302.3
2020-10-23	5	+0.1829	-32.5

A1.17. Source patient 65-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2475; median raw-area contrast +57.82. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 41. A1 source patient 65-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-15	0.008192	+0.1919	+15.4
2020-10-15	0.02048	+0.1633	+42.4
2020-10-15	0.0512	+0.2534	+101.4
2020-10-15	0.128	+0.1381	-37.9
2020-10-15	0.32	+0.1599	+61.3
2020-10-15	0.8	+0.1846	+4.7
2020-10-15	2	+0.2454	+94.4
2020-10-15	5	+0.2965	+69.6
2020-10-23	0.008192	+0.1632	-3.8
2020-10-23	0.02048	+0.2079	+87.6
2020-10-23	0.0512	+0.2214	+19.4
2020-10-23	0.128	+0.3708	+148.7
2020-10-23	0.32	+0.4989	+141.1
2020-10-23	0.8	+0.2196	-36.8
2020-10-23	2	+0.2531	+73.6
2020-10-23	5	+0.2677	+95.7
2020-11-05	0.008192	+0.1974	+23.7
2020-11-05	0.02048	+0.2906	+92.2
2020-11-05	0.0512	+0.1865	+38.7
2020-11-05	0.128	+0.3228	+154.6
2020-11-05	0.32	+0.2499	+54.3
2020-11-05	0.8	+0.4921	+62.0
2020-11-05	2	+0.2495	+10.8
2020-11-05	5	+0.2790	+9.3

A1.18. Source patient 66-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2951; median raw-area contrast +84.77. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 42. A1 source patient 66-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-15	0.008192	+0.1924	+12.4
2020-10-15	0.02048	+0.2020	+46.6
2020-10-15	0.0512	+0.2868	+56.2
2020-10-15	0.128	+0.3330	+36.3
2020-10-15	0.32	+0.2706	+94.5
2020-10-15	0.8	+0.1987	+63.7

2020-10-15	2	+0.1706	+45.6
2020-10-15	5	+0.3330	+68.4
2020-10-27	0.008192	+0.1634	+66.3
2020-10-27	0.02048	+0.2660	+90.9
2020-10-27	0.0512	+0.3031	+113.0
2020-10-27	0.128	+0.3836	+39.4
2020-10-27	0.32	+0.4641	+137.8
2020-10-27	0.8	+0.2784	+53.5
2020-10-27	2	+0.2882	+82.8
2020-10-27	5	+0.3812	+139.7
2020-11-05	0.008192	+0.1517	+105.6
2020-11-05	0.02048	+0.3261	+151.9
2020-11-05	0.0512	+0.3985	+114.8
2020-11-05	0.128	+0.3465	+14.0
2020-11-05	0.32	+0.4124	+154.4
2020-11-05	0.8	+0.3754	+192.8
2020-11-05	2	+0.3020	+100.9
2020-11-05	5	+0.1208	+86.7

A1.19. Source patient 68-CF

24 eligible of 24 blocks; median normalized treatment contrast +0.2833; median raw-area contrast +96.63. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 43. A1 source patient 68-CF: 24 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-21	0.008192	+0.1733	+76.5
2020-10-21	0.02048	+0.2356	+90.2
2020-10-21	0.0512	+0.3633	+172.9
2020-10-21	0.128	+0.4029	+17.3
2020-10-21	0.32	+0.3608	+120.1
2020-10-21	0.8	+0.3152	+205.5
2020-10-21	2	+0.2931	+103.5
2020-10-21	5	+0.3547	+103.1
2020-10-27	0.008192	+0.1400	+69.6
2020-10-27	0.02048	+0.1795	+125.8
2020-10-27	0.0512	+0.2906	+124.6
2020-10-27	0.128	+0.3525	+148.7
2020-10-27	0.32	+0.1431	+6.5
2020-10-27	0.8	+0.3659	+215.5

2020-10-27	2	+0.2594	+176.9
2020-10-27	5	-0.0718	-87.1
2020-11-27	0.008192	+0.1909	-4.2
2020-11-27	0.02048	+0.2265	+82.5
2020-11-27	0.0512	+0.3456	+52.2
2020-11-27	0.128	+0.4028	+153.4
2020-11-27	0.32	+0.2824	+58.8
2020-11-27	0.8	+0.2157	+68.5
2020-11-27	2	+0.1461	+27.8
2020-11-27	5	+0.2842	+156.9

A1.20. Source patient 69-CF

8 eligible of 8 blocks; median normalized treatment contrast +0.1718; median raw-area contrast +52.84. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 44. A1 source patient 69-CF: 8 plate-dose blocks from results/drevinek_fis.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
2020-10-23	0.008192	+0.1478	-6.2
2020-10-23	0.02048	+0.1696	+17.8
2020-10-23	0.0512	+0.1919	+47.8
2020-10-23	0.128	+0.2667	+81.7
2020-10-23	0.32	+0.1560	-23.6
2020-10-23	0.8	+0.1740	+139.8
2020-10-23	2	+0.2589	+57.9
2020-10-23	5	+0.0735	+80.4

A2. VTI minus ETI

Source <https://zenodo.org/records/15754800>. 12 IDs and 33 plates appear in the public source; 208/264 blocks meet its preregistered primary criteria. The table preserves positive and negative contrasts. The registered patient-level result is 6/9 positive medians, exact one-sided $p=0.2539062$. Unnormalized raw-area AUC is shown only to reveal sensitivity to baseline area, never as an independent trial.

A2.1. Source patient 06

24 eligible of 24 blocks; median normalized treatment contrast +0.0277; median raw-area contrast -8.49. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 45. A2 source patient 06: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240409	0.008192	+0.0673	+14.6
20240409	0.02048	+0.1224	+125.7
20240409	0.0512	+0.1283	-8.1
20240409	0.128	+0.0963	+56.1
20240409	0.32	-0.2065	+39.9
20240409	0.8	-0.0746	-33.6
20240409	2	+0.0106	+119.2
20240409	5	-0.1393	-49.6
20240416	0.008192	+0.1074	-24.7
20240416	0.02048	+0.0635	+80.1
20240416	0.0512	-0.0403	+16.7
20240416	0.128	+0.0839	-19.4
20240416	0.32	-0.0454	-26.3
20240416	0.8	+0.0135	-8.8
20240416	2	-0.1746	-47.1
20240416	5	-0.0725	-24.5
20240430	0.008192	+0.0397	-30.6
20240430	0.02048	+0.0156	+38.1
20240430	0.0512	+0.0455	-112.0
20240430	0.128	+0.0783	-134.1
20240430	0.32	-0.0534	+50.5
20240430	0.8	-0.0909	+29.6
20240430	2	+0.1366	-110.9
20240430	5	+0.2357	+135.0

A2.2. Source patient 27

24 eligible of 24 blocks; median normalized treatment contrast +0.0449; median raw-area contrast -4.67. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 46. A2 source patient 27: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240409	0.008192	+0.1404	+19.1
20240409	0.02048	+0.1543	+20.4
20240409	0.0512	+0.1201	+8.6

20240409	0.128	+0.2176	+48.1
20240409	0.32	-0.0955	-56.9
20240409	0.8	-0.0967	-102.1
20240409	2	-0.1990	+33.4
20240409	5	-0.0490	-5.1
20240416	0.008192	+0.0569	-1.1
20240416	0.02048	+0.0256	-13.5
20240416	0.0512	-0.0417	-40.9
20240416	0.128	+0.1041	-55.2
20240416	0.32	+0.0610	-17.7
20240416	0.8	-0.1142	+19.5
20240416	2	+0.1909	-4.2
20240416	5	-0.0018	-53.3
20240430	0.008192	+0.0236	+123.2
20240430	0.02048	-0.0882	+35.6
20240430	0.0512	+0.0329	-43.1
20240430	0.128	+0.1594	-49.5
20240430	0.32	+0.1445	-29.3
20240430	0.8	+0.2277	+26.0
20240430	2	-0.0951	-138.9
20240430	5	+0.0678	+96.4

A2.3. Source patient 31

24 eligible of 24 blocks; median normalized treatment contrast -0.0071; median raw-area contrast +1.27. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 47. A2 source patient 31: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240517	0.008192	+0.0205	+56.5
20240517	0.02048	-0.0042	+46.6
20240517	0.0512	-0.0206	-9.6
20240517	0.128	-0.0734	+20.9
20240517	0.32	-0.0520	-86.1
20240517	0.8	-0.0064	+26.0
20240517	2	-0.0664	-11.4
20240517	5	+0.0476	+23.2
20240523	0.008192	-0.0078	+6.4
20240523	0.02048	+0.0407	-95.6
20240523	0.0512	+0.0226	+136.1

20240523	0.128	-0.0575	-30.9
20240523	0.32	+0.0873	-109.2
20240523	0.8	-0.0168	+74.6
20240523	2	+0.2221	+129.3
20240523	5	-0.0983	-144.8
20240530	0.008192	-0.0413	-0.1
20240530	0.02048	+0.0166	-1.0
20240530	0.0512	-0.0805	-21.8
20240530	0.128	+0.0045	+2.6
20240530	0.32	+0.0143	-10.7
20240530	0.8	-0.0503	+54.3
20240530	2	+0.0192	+28.2
20240530	5	-0.2074	-43.3

A2.4. Source patient 35

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 48. A2 source patient 35: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20240326	0.008192	excluded	missing matched times
20240326	0.02048	excluded	missing matched times
20240326	0.0512	excluded	missing matched

			times
20240326	0.128	excluded	missing matched times
20240326	0.32	excluded	missing matched times
20240326	0.8	excluded	missing matched times
20240326	2	excluded	missing matched times
20240326	5	excluded	missing matched times

A2.5. Source patient 40

16 eligible of 24 blocks; median normalized treatment contrast -0.0074; median raw-area contrast +3.17. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 49. A2 source patient 40: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20230627	0.008192	excluded	missing matched times
20230627	0.02048	excluded	missing matched times
20230627	0.0512	excluded	missing matched times
20230627	0.128	excluded	missing matched times
20230627	0.32	excluded	missing matched times
20230627	0.8	excluded	missing matched times
20230627	2	excluded	missing matched times
20230627	5	excluded	missing matched times
20241121	0.008192	+0.0078	-21.6
20241121	0.02048	+0.0141	-38.6
20241121	0.0512	-0.1030	+22.9
20241121	0.128	-0.0426	-146.0
20241121	0.32	+0.2843	-188.0
20241121	0.8	-0.3458	-179.2

20241121	2	-0.0569	-260.8
20241121	5	+0.1733	+14.7
20241204	0.008192	+0.0137	+13.7
20241204	0.02048	+0.0022	+20.5
20241204	0.0512	-0.0329	+74.1
20241204	0.128	-0.0665	-165.2
20241204	0.32	-0.0024	+71.5
20241204	0.8	-0.0125	+286.5
20241204	2	-0.0851	+134.1
20241204	5	+0.0549	-7.3

A2.6. Source patient 45

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 50. A2 source patient 45: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20240328	0.008192	excluded	missing matched times
20240328	0.02048	excluded	missing matched times
20240328	0.0512	excluded	missing matched times
20240328	0.128	excluded	missing matched times

20240328	0.32	excluded	missing matched times
20240328	0.8	excluded	missing matched times
20240328	2	excluded	missing matched times
20240328	5	excluded	missing matched times

A2.7. Source patient 46

24 eligible of 24 blocks; median normalized treatment contrast +0.0345; median raw-area contrast -2.46. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 51. A2 source patient 46: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240523	0.008192	-0.0261	-1.5
20240523	0.02048	-0.0601	+35.9
20240523	0.0512	+0.0294	-3.4
20240523	0.128	+0.1802	+39.0
20240523	0.32	-0.1560	-95.0
20240523	0.8	+0.0617	+59.8
20240523	2	-0.1395	+5.8
20240523	5	+0.1327	+28.1
20240530	0.008192	+0.0655	-28.3
20240530	0.02048	-0.0300	-21.6
20240530	0.0512	+0.1325	+76.6
20240530	0.128	+0.0425	-12.1
20240530	0.32	+0.1158	-20.2
20240530	0.8	-0.1137	+82.9
20240530	2	+0.0750	+129.5
20240530	5	-0.0437	-38.0
20240625	0.008192	+0.0610	+126.1
20240625	0.02048	+0.0397	-132.9
20240625	0.0512	+0.1187	-141.3
20240625	0.128	-0.1928	-28.4
20240625	0.32	-0.1743	-94.4
20240625	0.8	-0.1692	-365.7
20240625	2	+0.0176	+43.8
20240625	5	+0.3418	+282.1

A2.8. Source patient 58

No eligible primary block for this ID under the fixed treatment arms and time/well criteria. 16 source blocks are retained below as exclusions; this ID contributes no patient median or sign-test observation.

Table 52. A2 source patient 58: 16 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240314	0.008192	excluded	missing matched times
20240314	0.02048	excluded	missing matched times
20240314	0.0512	excluded	missing matched times
20240314	0.128	excluded	missing matched times
20240314	0.32	excluded	missing matched times
20240314	0.8	excluded	missing matched times
20240314	2	excluded	missing matched times
20240314	5	excluded	missing matched times
20241125	0.008192	excluded	missing matched times
20241125	0.02048	excluded	missing matched times
20241125	0.0512	excluded	missing matched times
20241125	0.128	excluded	missing matched times
20241125	0.32	excluded	missing matched times
20241125	0.8	excluded	missing matched times
20241125	2	excluded	missing matched times
20241125	5	excluded	missing matched times

A2.9. Source patient 68

24 eligible of 24 blocks; median normalized treatment contrast -0.0086; median raw-area contrast -6.57. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 53. A2 source patient 68: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240523	0.008192	+0.1833	+10.7
20240523	0.02048	-0.0085	-19.6
20240523	0.0512	-0.0683	+104.4
20240523	0.128	-0.0599	+6.4
20240523	0.32	-0.0783	+32.1
20240523	0.8	-0.1295	-111.1
20240523	2	-0.0366	+31.4
20240523	5	-0.0529	-222.3
20240618	0.008192	+0.0920	-164.7
20240618	0.02048	+0.0920	+34.0
20240618	0.0512	-0.0680	-83.5
20240618	0.128	+0.2306	+205.8
20240618	0.32	+0.0186	+15.5
20240618	0.8	+0.0648	+58.2
20240618	2	-0.0087	+118.5
20240618	5	+0.0629	+198.5
20240625	0.008192	+0.0275	-111.6
20240625	0.02048	+0.1288	-50.9
20240625	0.0512	+0.0323	-112.7
20240625	0.128	+0.0999	+86.8
20240625	0.32	-0.1190	-175.5
20240625	0.8	-0.0566	-151.4
20240625	2	-0.3497	-407.8
20240625	5	-0.1398	-215.7

A2.10. Source patient 91

24 eligible of 24 blocks; median normalized treatment contrast +0.0282; median raw-area contrast -18.70. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 54. A2 source patient 91: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240411	0.008192	+0.0229	-6.6
20240411	0.02048	-0.0303	-14.2
20240411	0.0512	-0.1336	-142.0
20240411	0.128	+0.0830	-23.6
20240411	0.32	-0.0511	-60.4
20240411	0.8	-0.0090	-48.8
20240411	2	+0.1982	+28.7
20240411	5	-0.0349	-25.5
20240418	0.008192	+0.0667	+4.1
20240418	0.02048	+0.0713	+34.1
20240418	0.0512	-0.0999	+44.8
20240418	0.128	+0.1409	-7.5
20240418	0.32	-0.0271	-12.9
20240418	0.8	-0.3972	-113.6
20240418	2	-0.1867	-130.8
20240418	5	+0.2601	+74.7
20240530	0.008192	+0.0539	-58.3
20240530	0.02048	+0.0612	-89.6
20240530	0.0512	+0.2280	+88.2
20240530	0.128	-0.0384	-79.0
20240530	0.32	+0.0473	-23.2
20240530	0.8	+0.0482	+14.0
20240530	2	+0.0335	+87.7
20240530	5	-0.0604	-64.9

A2.11. Source patient 94

24 eligible of 24 blocks; median normalized treatment contrast +0.0181; median raw-area contrast -1.68. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 55. A2 source patient 94: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240517	0.008192	-0.0207	+32.3
20240517	0.02048	+0.1247	-15.3
20240517	0.0512	+0.1258	+37.5
20240517	0.128	+0.0476	+63.3
20240517	0.32	-0.0145	-81.3
20240517	0.8	+0.0029	-5.6

20240517	2	+0.0201	-212.3
20240517	5	-0.2070	-9.6
20240523	0.008192	+0.0776	-5.9
20240523	0.02048	-0.0184	+2.2
20240523	0.0512	-0.0364	+111.8
20240523	0.128	+0.0162	-11.8
20240523	0.32	-0.1115	+2.4
20240523	0.8	-0.0523	+17.8
20240523	2	-0.0246	+79.5
20240523	5	+0.0761	-146.5
20240618	0.008192	+0.0082	+151.4
20240618	0.02048	+0.1450	-247.8
20240618	0.0512	+0.0795	+27.1
20240618	0.128	+0.1825	+126.8
20240618	0.32	+0.0732	-34.9
20240618	0.8	-0.0403	-20.6
20240618	2	+0.0750	+165.2
20240618	5	+0.0522	-131.9

A2.12. Source patient 96

24 eligible of 24 blocks; median normalized treatment contrast +0.0039; median raw-area contrast +14.80. This is one patient-weighted observation regardless of the number of doses or plates. A negative or small block remains in the table rather than being filtered by outcome.

Table 56. A2 source patient 96: 24 plate-dose blocks from results/vti_eti.json. Raw-area delta is not baseline normalized.

plate date	fsk (uM)	normalized delta	raw delta or exclusion
20240510	0.008192	-0.0213	+22.4
20240510	0.02048	-0.0038	+36.6
20240510	0.0512	+0.0183	-58.3
20240510	0.128	+0.1031	+11.8
20240510	0.32	-0.0187	-26.2
20240510	0.8	+0.1301	+35.0
20240510	2	+0.0947	+88.4
20240510	5	-0.0399	+22.1
20240517	0.008192	+0.0189	+20.4
20240517	0.02048	-0.0141	-19.0
20240517	0.0512	+0.0410	+2.9
20240517	0.128	-0.0070	-30.7
20240517	0.32	-0.0814	+95.5
20240517	0.8	-0.1982	+129.7

20240517	2	+0.0774	+17.8
20240517	5	+0.0117	+9.6
20240620	0.008192	+0.1536	+5.1
20240620	0.02048	+0.0341	-50.5
20240620	0.0512	+0.1347	+79.0
20240620	0.128	+0.0341	-47.5
20240620	0.32	-0.1029	+18.5
20240620	0.8	-0.2127	-150.3
20240620	2	-0.0377	-122.1
20240620	5	-0.1185	+44.4

Audit boundary. A plate-dose block is not an independent donor. Within each patient the median across eligible blocks gives one observation for the exact sign test, preventing a large plate or many concentrations from dominating the patient count. Raw-area reversals flag sensitivity to baseline organoid density. For excluded A2 patient blocks, a missing default treatment arm is not replaced with a different VTI concentration. This appendix is a reproducibility trace of two already published in-vitro comparisons, not evidence for clinical benefit, equivalence, or the single-object size candidate.